



## Historical Perspective

Mineral precipitation in porous media systems: Controlling factors and impacts on porous media evolution<sup>☆</sup>Hang Deng<sup>a,b,c,\*</sup>, Jenna Poonoosamy<sup>d</sup><sup>a</sup> Department of Energy and Resources Engineering, Peking University, Beijing 100871, China<sup>b</sup> Institute of Energy, Peking University, Beijing, 100871, China<sup>c</sup> Institute of Carbon Neutrality, Peking University, Beijing, 100871, China<sup>d</sup> Institute of Fusion Energy and Nuclear Waste Management (IFN-2): Nuclear Waste Management, Forschungszentrum Jülich GmbH, 52425 Jülich, Germany

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## ABSTRACT

Mineral precipitation in porous media is an important process that controls the evolution of physical-chemical properties of many natural and engineered systems. This review aims to provide an overview of the status quo understandings of this process and a perspective on future research needs. We reviewed the fundamentals of the thermodynamic and kinetic theories of nucleation and crystal growth, and the key factors (e.g., substrate properties and pore sizes) that control mineral precipitation in porous media. In particular, we focused on how local reactive transport conditions dictate the extent and spatial distribution of mineral precipitates in flowing supersaturated solutions, mixing dominated systems, and coupled mineral dissolution and precipitation systems. Moreover, the review summarizes constitutive relations that have been used to model precipitation-driven alteration in bulk properties such as permeability, diffusivity and mineral reactivity, as well as their applicability and limitations. While there have been exciting developments in experimental and modeling efforts in improved understanding, some outstanding research questions require dedicated future investigations. They include more rigorous descriptions of emerging precipitation phenomena such as oscillatory zoning during the precipitation of solid solutions, considerations of the coupling between geochemical and geomechanical processes with precipitation induced cracking being a worthy example, and the upscaling of fine scale process and properties (e.g., confinement, multiphase flow dynamics) in to macroscopic predictive models. Overall, this review highlights the importance, advances and challenges of improved understanding of mineral precipitation in fractured porous media.

## 1. Introduction

Mineral precipitation is a widely observed phenomenon in natural and engineered systems. It is a major reaction during chemical weathering [44], and can provide valuable records of the environmental conditions (e.g., the paleoclimate) [178]. It is a key process that shapes the global biogeochemical cycling of e.g., Fe [102] and magmatic-hydrothermal ore formation [57]. Deeper understandings of mineral precipitation also have important implications for medical studies of dental repair [88], kidney stone formation [163], etc. In a multitude of engineered systems, mineral precipitation is a useful technique. Coprecipitation has been a long-practiced technique to remove contaminants (e.g., As, Cd, sulfate) [22,145] for water treatment or to recover

resources (e.g., phosphorous) from waste water and sewage sludge [25,60,185]. Given the urgent need of mitigating climate change, deployment of technologies that heavily rely on CO<sub>2</sub> mineralization for carbon removal and storage makes the studies of mineral precipitation of particular interest and importance [26,32,75,76,99–101,108,134,165]. In the conventional geologic carbon storage systems, e.g., saline aquifers, mineralization trapping is the most secure storage mechanism that dominates on the time scale of hundreds and thousands of years [103]; whereas in unconventional storage systems such as basaltic formations, mineralization can occur within years [99,100]. Enhanced rock weathering [10] and carbonation of alkaline solid wastes [135,136] are technologies that have been actively explored for the purpose of carbon removal and other co-benefits such as improving soil

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health and achieving carbon neutrality for the mining industry. Mineral precipitation, especially microbial induced, provides a means to seal flow pathways to prevent contaminant migration [27] or leakage from target reservoirs for geologic carbon storage [19,125] and nuclear waste disposal [2]. Mineral precipitation, however, may be undesired and thus needs to be managed in certain engineered systems. In hydraulic fracturing systems of unconventional oil/gas reservoirs [11,193] or geothermal systems [68], mineral precipitates in the flow paths can have negative effects on energy production.

Mineral precipitation occurs when the solution becomes supersaturated with respect to the mineral phases. The changes in the saturation state are commonly associated with concentration evolution. The loss of water is one of the mechanisms that result in solute concentration increase and mineral precipitation, which can occur in geothermal systems involving boiling and cooling [57,68] and in geologic carbon storage or CO<sub>2</sub>-EOR (enhanced oil recovery) systems when water dissolves in the supercritical CO<sub>2</sub> phase [26,45,53]. Chemical reactions including mineral dissolution and oxidation-reduction can also trigger mineral precipitation by providing the solutes that are also the reactants for the precipitation reaction [141,194]. One example is basalt mineralization, which involves the dissolution of Ca and Mg rich mineral phases that releases cations in carbonate rich solutions, leading to the carbonation reaction [76,134]. The saturation state may increase as a result of decreases in mineral solubility, which is the case in geothermal systems when the solution at equilibrium with rocks at one temperature migrates to locations at temperatures of lower solubility [119]. Fluid migration plays a critical role in determining local concentrations and thus saturation states. Mixing of two solutions of different compositions (e.g., groundwater and seawater, waterflooding) or temperatures may also bring up the saturation state of the mixture directly or change the conditions (e.g., redox) that eventually lead to over-saturation indirectly [17,53,208]. Microbes, through their metabolic activities and presence of cell surfaces, may create a local environment that is super-saturated and thereby induce and facilitate mineral precipitation [58].

Mineral precipitation in fractured porous media is a research area of particular importance and necessity [34]. The structures of fractured porous media dictate fluid flow and solute transport in natural and engineered systems, which in turn influence local concentrations of different chemical species and thus the spatial distribution and extent of mineral dissolution and precipitation [50,84,111,112]. Mineral precipitation can alter the structures by reducing pore/aperture sizes and clogging pore throats [13,16,96,200]. Consequently, hydrodynamics and transport are modified, resulting in pressure oscillations or re-distributions of reaction fronts [16,162]. This intricate coupling between mineral precipitation, flow and transport will largely affect the long-term evolution of the fractured porous media and their chemical-physical properties [46,50,87,94]. At the continuum scale, properties such as porosity, permeability, diffusivity, and mineral reactivity evolve in response to mineral precipitation, and the trajectories can deviate from commonly used constitutive relations (e.g., the Kozeny Carman relation and Archie's law) and have different characteristics from those developed for capturing alterations driven by mineral dissolution [20,35,128].

Therefore, improved understandings of mineral precipitation in fractured porous media and the resulting alteration are essential for better assessment of many natural and engineered systems. In this article, we aim to provide a review of recent advances on this topic and a perspective on the outstanding challenges and the outlook for future research. We start with an overview of the fundamentals of the thermodynamics and kinetics of mineral precipitation (Section 2), followed by a discussion of the factors that control mineral precipitation in fractured porous media (Section 3) and the impacts of mineral precipitation on the physical-chemical properties of fractured porous media (Section 4). We will then present recent development focusing on improved pore-scale insights from experimental and modeling efforts, with some thoughts on the frontiers of mineral precipitation in porous

media (Section 5), and end with a few concluding remarks (Section 6).

## 2. Fundamentals of mineral precipitation

In this section, we provide an overview of the thermodynamics and kinetics of mineral precipitation. The thermodynamics govern whether a mineral phase will be created, whereas the kinetics determine how fast the mineral phase will be generated. Mineral precipitation typically includes two processes: Nucleation and crystal growth. Nucleation refers to the initial stage of crystal formation from the solution or on a different mineral phase, and crystal growth is the continued growth of the existing crystal. On a macroscopic scale, nucleation and crystal growth may occur at the same time, and collectively control the precipitation rate.

### 2.1. Thermodynamics of mineral precipitation

Mineral precipitates when it is over-saturated in the solution. The saturation state for a mineral reaction as given in Eq. (1) can be quantified by the saturation ratio ( $\Omega$ ), i.e., the ratio between ionic activity product (IAP, Eq. (2)) and the solubility coefficient ( $K_{sp}$ ), or the saturation index (SI, Eq. (3)).

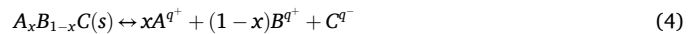


$$IAP = a_A^{q+} \cdot a_C^{q-} \quad (2)$$

$$SI = \ln\left(\frac{IAP}{K_{sp}}\right) = \ln(\Omega) = \Delta G / 2.3RT \quad (3)$$

where  $a_i$  is the activity of species  $i$ , and  $\Delta G$  is the Gibbs free energy of the reaction,  $R$  is ideal gas constant, and  $T$  is the absolute temperature. Precipitation occurs when  $SI > 0$ , i.e., the reversed reaction of Eq. (1) is thermodynamically favored.

In reality, more than one mineral phases may co-exist as a mixture, aka solid solution. For an ideal solid solution defined as.



the solubility constant can be calculated by Eq. (5a).

$$K_x = K_{AC}^x K_{BC}^{1-x} x^x (1-x)^{1-x} \quad (5a)$$

$$K_x = K_{AC}^x \gamma_{AC}^x x^x K_{BC}^{1-x} \gamma_{BC}^{1-x} (1-x)^{1-x} \quad (5b)$$

where  $x$  is the fraction of phase AC, and  $K_{AC}, K_{BC}$  are the solubilities of the two end members. For non-ideal mixing, the activity of the end members (i.e.,  $\gamma_{AC}, \gamma_{AB}$ ) in the solid solution needs to be accounted for (Eq. (5b)) [130,133].

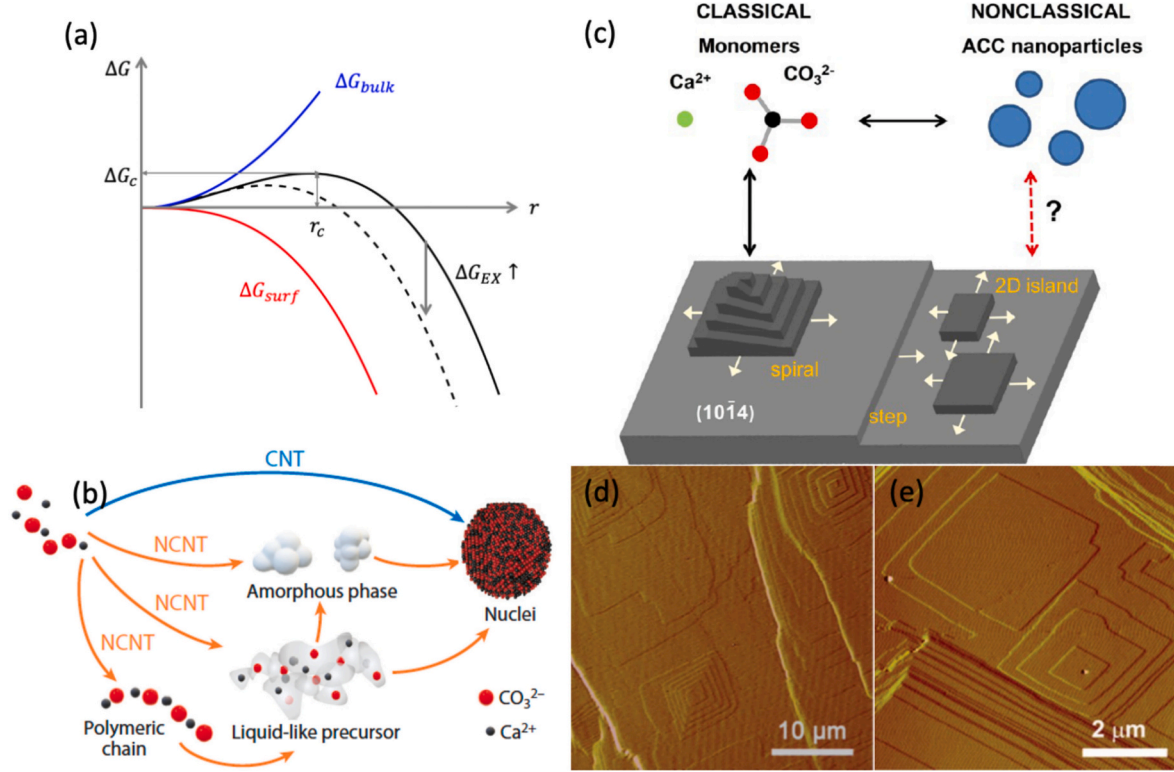
### 2.2. Kinetics of mineral precipitation

The other important factor that determines whether precipitation can be detected within an observational timeframe is the kinetics of the precipitation reaction. Here, we present the theoretical models used to mathematically describe nucleation and crystal growth.

#### 2.2.1. Nucleation rate

Classical Nucleation Theory (CNT) provides a useful theoretical framework for quantitative discussions of the nucleation process. Based on the Classical Nucleation Theory, the Gibbs free energy of a nucleus is the summation of its bulk free energy and the surface energy, which can be formulated as follows assuming a spherical shape (Fig. 1a).

$$\Delta G = -\frac{4\pi r^3}{3v_0} k_B T \ln \Omega + 4\pi r^2 \sigma \quad (6)$$



**Fig. 1.** (a) Gibbs free energy as a function of the radius of the nucleus following the classical nucleation theory, and the changes due to the excess Gibbs free energy associated with pre-nucleation clusters as proposed by [52]; (b) illustration of the classical and non-classical pathways [72]; (c) classical and non-classical nucleation on calcite [153]; and the AFM images of (d) spiral growth and (e) 2D nucleation on calcite [1014] surfaces [156].

where  $r$  is the radius of the nucleus,  $v_0$  is the volume of a single molecule,  $k_B$  is the Boltzmann constant, and  $\sigma$  is the interfacial energy. As the bulk free energy decreases (i.e., is more negative) as the radius increases while the free energy increases, a critical radius exists when the Gibbs free energy is maximum and the probability of formation of the nucleus is minimum. The critical radius ( $r_c$ ) can be found by setting the first-order derivative of Eq. (6) to zero,

$$r_c = \frac{2\sigma v_0}{k_B T \ln \Omega} \quad (7)$$

and substituting it back into (6) gives the corresponding Gibbs free energy as:

$$\Delta G_c = \frac{B v_0^2 \sigma^3}{(k_B T \ln \Omega)^2} \quad (8)$$

where  $B$  is the geometric factor and is  $\frac{16\pi}{3}$  for spherical nuclei, and can be adjusted for other shapes. The nucleation rate ( $J$ ) is then given by:

$$J = \Gamma \exp\left(-\frac{\Delta G_c}{k_B T}\right) \quad (9)$$

where  $\Gamma$  is the pre-exponential term, and can be calculated from the nucleation sites density ( $N_0$ ), the number of monomers per unit volume of fluid ( $N_1$ , [# monomers/ $m^3$  fluid]), and the diffusion coefficient of the monomer ( $D$ ) (Eq. (10)). Typically,  $N_0$  has a unit of [# nuclei/ $m^2$  substrate] for heterogeneous nucleation, i.e., nucleation on a substrate, and a unit of [# nuclei/ $m^3$  solution] for homogeneous nucleation, i.e., nucleation in a solution.  $Z$  in Eq. (10) is a function of the Gibbs free energy and the number of the critical nuclei ( $n_c$ ) (Eq. (11)), and  $a$  in Eq. (12) is the area of the nuclei [140].

$$\Gamma = 4\pi Z D N_0 N_1 r_c \quad (10)$$

$$Z = \sqrt{\left(\frac{\Delta G_c}{3\pi k_B T (n_c)^2}\right)} \quad (11)$$

$$n_c = \left(\frac{2\sigma a}{3k_B T \ln \Omega}\right)^3 \quad (12)$$

For a given mineral system, extra efforts are typically required to parameterize interfacial tension ( $\sigma$ ), and the pre-exponential term ( $\Gamma$ ) in Eq. (9) or the nucleation site density ( $N_0$ ) in Eq. (10); whereas other parameters can be more readily determined based on fluid chemistry (e.g.,  $\ln \Omega$  and  $N_1$ ) and the basic information of the mineral phase (e.g.,  $v_0$ ). Because  $J$  can be resolved experimentally by tracking the number of nuclei, and its log-value ( $\ln J$ ) and  $(\ln \Omega)^2$  follow a linear relationship,  $\ln \Gamma$  the  $\sigma$  can be determined from the fitted intercept and slope according to Eq. (8) & (9).  $N_0$  can then be determined from Eq. (10). Experimental approach varies based on how the number of nuclei is monitored (Table 1). Techniques including optical microscopic measurements [47,204], the Grazing Incidence Small-Angle X-Ray Scattering [41], nano-scale computed tomography imaging [204], and Atomic Force Microscopy [154] have been employed. Furthermore, the effective interfacial tension for heterogeneous nucleation ( $\sigma_{het}$ ) of the precipitating mineral phase on the substrate can also be calculated from the interfacial tension between the precipitating phase and the solution, i.e., for homogeneous nucleation ( $\sigma_{homo}$ ) [74]. For any contact angle ( $\theta$ ),  $\sigma_{het} \leq \sigma_{homo}$ , which is consistent with the observation that homogeneous nucleation requires a higher energy barrier be exceeded.

$$\sigma_{het} = \left(\frac{1}{4}(2 + \cos \theta)(1 - \cos \theta)^2\right)^{1/3} \sigma_{homo} \quad (13)$$

However, the nucleation process of some minerals does not always follow the classical pathway, for which the nuclei form through addition of the monomers. The non-classical pathway has been widely reported

**Table 1**

Determination of the key parameters of the kinetic models for mineral precipitation.

| Kinetic model                                                 | Equations                  | Key parameters       | Determination of the key parameters                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------|----------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classical Nucleation Theory                                   | Eqs. (8)–(9)/Eqs. (9)–(12) | $\Gamma/N_0, \sigma$ | Determined by fitting the intercept and slope of $\ln J = \ln \Gamma + \left( -\frac{B\sigma^3 v_0}{2.3^2 (kT)^3} \right) \frac{1}{SI^2}$ , $J$ is monitored experimentally using methods including:<br>Optical microscopy [47,204]<br>Spectroscopic method: Gazing Incidence Small-angle X-ray Scattering [41]<br>Nano computed tomography imaging [204]<br>Atomic Force Microscopy [154] |
|                                                               |                            |                      | Determined from the progressions of reaction that can be monitored via:<br>Stirred batch experiments [7,62,115]<br>Computed tomography imaging [115,116,175]<br>Atomic Force Microscopy [175]                                                                                                                                                                                              |
| Transition State Theory for crystal growth/precipitation rate | Eqs. (14)/(15)/(16)        | $k_{rxn}$            |                                                                                                                                                                                                                                                                                                                                                                                            |

for minerals like calcite, which involves multistages and intermediate species, such as prenucleation clusters, dense liquid phases, and amorphous phases [31,72,164] (Fig. 1b). Cautions need to be exercised when using the CNT formulations to model these minerals, even though it has been suggested that each step involved in the non-classical pathway may be described by the CNT [164]. One approach that has been proposed to bridge the CNT formulation and the non-classical nucleation process is to add an excess free energy term that is associated with the interfacial energy of the prenucleation clusters to captured the lowering of the energy barrier [52] (Fig. 1a).

### 2.2.2. Crystal growth rate

Mineral precipitation is typically modeled using the Transition State Theory (TST) type of reaction law, which has been adopted as it provides a unified theoretical framework to relate kinetics with thermodynamics [82,160]. The basic formulation for an elementary reaction is given in Eq. (14) and the rate is negative for mineral precipitation.

$$R = A \bullet k_{rxn} \bullet \left( 1 - e^{\frac{\Delta G}{RT}} \right) = A \bullet k_{rxn} \bullet (1 - \Omega) \quad (14)$$

where  $A$  is the surface area, and  $k_{rxn}$  is the intrinsic kinetic coefficient. The chemical affinity term describes the thermodynamic driving force or the distance to equilibrium.

The key parameter of Eq. (14) is  $k_{rxn}$ , which can be determined via stirred batch reactor experiments, similar to the determination of the kinetic coefficient for mineral dissolution reactions. In order to minimize the influence of nucleation on the assessment of the kinetic coefficient for crystal growth, the batch experiments are typically seeded and the supersaturation is managed at a level when homogeneous nucleation is suppressed [7,62,115]. Surface area of the solid phase can be assessed through BET measurement and fluid chemistry over time provides reaction progress and the chemical affinity, which then can be substituted into Eq. (14) to calculate  $k_{rxn}$ . Palandri [120] and the references therein provide a comprehensive compilation of the kinetic data for mineral

dissolution and precipitation and the protocols used for their determination. Image-based methods can also be used to monitor directly surface area and the amount of reaction (Table 1), and thus to evaluate  $k_{rxn}$ . For example, Computed Tomography [115,116] and Atomic Force Microscopy [141,175] can record time-resolved micron-meter scale 3D crystal morphology and nano-meter scale 2D surface topography, respectively. While  $k_{rxn}$  can be calculated using Eq. (14) from CT data directly, it is typically calculated from the advancing velocity at the step edges (Fig. 1c) from AFM data. However, the challenge with these methods compared to the stirred experiments is the transport limitation at the mineral interfaces [115], which will be discussed in more details in Section 3.3.3.

### 2.2.3. Combined kinetic rate

It should be noted that although nucleation has to proceed crystal growth in solutions, the two processes do not necessarily occur in a sequential order. The relative contribution of nucleation versus growth is dependent on the local saturation state and the evolution of the substrate surface. Therefore, the TST type of rate law has also been used to simulate the precipitation process as a whole in addition to crystal growth. The simplified general form is typically used to fit the experimental data (Eq. (15)). The reaction order ( $n$ ) is correlated with the reaction mechanisms (i.e., 2D nucleation growth versus spiral growth, Fig. 1c-e), and is a fitting parameter that may vary with the range of SI [175]. AFM studies have revealed that the 2D nucleation growth is the dominant mechanism at high SI and typically results in  $n > 2$ , while spiral growth on the dislocations typically result in  $1 < n < 2$  [175].

$$R = A k_{rxn} (1 - \Omega)^n \quad (15)$$

Precipitation rate has also been shown to vary with other aspects of fluid chemistry in addition to the thermodynamic driving force. For example, at the same supersaturation, varying the ratio between the cations and anions can result in different precipitation rates. A cation/anion ratio larger than one has been shown to promote nucleation and crystal growth of barite [80], whereas the highest precipitation rate of calcite was observed for a cation/anion ratio of one [122]. Additionally, pH and composition of the fluid may also affect the precipitation process by changing the local solvation environment or causing crystallographic distortion when incorporated [154,155,179].

The general form of the TST rate law enables considerations of the correlation between mineral reaction rates with fluid chemistry, by including the activity of any catalytic and inhibitory species ( $a_i$ ).

$$R = A \bullet k_{rxn} \bullet \exp\left(-\frac{E_a}{RT}\right) \bullet \prod_i a_i \bullet \left(1 - \left(\frac{IAP}{K_{sp}}\right)^{n_1}\right)^{n_2} \quad (16)$$

The temperature ( $T$ ) dependence is captured by the Arrhenius term  $\exp\left(-\frac{E_a}{RT}\right)$ , where  $E_a$  is the activation energy, and  $n_1$  and  $n_2$  are exponents that are determined by the reaction mechanisms.

## 3. Controlling factors of mineral precipitation in porous media

The geochemical factors discussed in Section 2 control the extent and rate of mineral precipitation, and more factors come into play in influencing the rate and spatial distributions of precipitation in porous media, including the properties of the substrate, the pore size distributions, and spatially variable flow and transport that affects local geochemical conditions.

### 3.1. Substrate

From a theoretical perspective, the substrate can affect the precipitation process by dictating the interfacial energy between the precipitating phase and the substrate (Eq. (8)). A lower interfacial energy, which might be caused by a higher degree of similarity in the



crystallographic structures between the precipitating phase and the substrate, would result in a faster nucleation and growth. The effect of substrate has been demonstrated experimentally. For example, in the study of Noiri et al. [115] and Noiri et al. [116], calcium carbonate precipitation was investigated by injecting supersaturated solutions into columns packed with a mixture of glass beads and a calcium carbonate phase, i.e., calcite spar or aragonite ooids (Fig. 2). Calcite spars were observed to be fully covered by epitaxial growth of the neoformed carbonate phase (Fig. 2a), aragonite grains were partially covered by polyhedral crystal aggregates, and glass beads were sparsely covered by relatively large rhombohedral crystals (Fig. 2b). Experiments of  $\text{CaCO}_3$  precipitation in fractures have also shown preferential precipitation in regions initially seeded with  $\text{CaCO}_3$  [69,113].

In addition to mineralogy, the surface topography of the substrate may also be important. In the counter-diffusion experiments of Leclerc et al. [83] that examined barite precipitation in carbonate rocks, aggregates of fine barite crystals were found in contact with micrites, whereas euhedral barite crystals formed on euhedral calcite. This difference was attributed to higher roughness and nucleation site density on the micrites.

The substrate can also affect the precipitation reaction by modifying local fluid chemistry. This is a dominant mechanism in microbial induced carbonate precipitation [58], and can also occur in abiotic systems. For example, clay minerals can absorb and desorb cations, and thus elevate local supersaturation that leads to fast formation of smaller nuclei, resulting in aggregates of small crystals around clay minerals [83]. This mechanism may affect not only heterogeneous nucleation and crystal growth, but also homogeneous nucleation.

In addition, mineral surfaces exhibit spatially variable reactivity due to the distribution of distinct surface sites such as kinks, steps, and terraces [3,14,95]. These features influence local dissolution rates and can lead to non-uniform secondary mineral growth. Studies on calcite and barite have demonstrated that dissolution is strongly controlled by crystallographic faces, etch pits, and kink sites [42], as well as by the geometry of these features [81]. Etch pits nucleate at defect or step edges and propagate, exposing low-coordination kink and edge sites that dissolve preferentially, enhancing the overall dissolution rate. Face orientation is also critical: the calcite [104] face dominates dissolution under typical experimental conditions because it exposes a higher density of reactive step and kink sites. AFM experiments have also reported different growth rates at the acute step versus the obtuse step of calcite [157]. Understanding surface heterogeneity is particularly relevant for interface-coupled mineral dissolution and precipitation because the presence of dissolved cations or anions can “poison” reactive sites and

inhibit further dissolution and prevent secondary mineral precipitation despite supersaturation. For example, studies of calcite dissolution under acidic conditions in the presence of  $\text{Pb}^{2+}$  have reported decreasing dissolution rates with increasing lead content [1]. These observations indicate that impurity incorporation can alter local reactivity and potentially inhibit secondary mineral formation, e.g., cerussite ( $\text{PbCO}_3$ ) in this case.

In multiphase systems, wettability of the substrate can also influence the precipitation process by shaping the multiphase hydrodynamics [105]. Microfluidic experiments of salt precipitation in presence of  $\text{CO}_2$  showed that for hydrophilic and neutral surfaces, salt precipitation can occur outside of the initial residual brine regions, indicating migration of the brine in the process, and salt precipitation occurs at the locations of the initial residual brine when the surfaces are hydrophobic [56]. Interfacial area between water and gas or between water and the mineral phases have also been reported to be important for other mineral reactions in unsaturated systems [54].

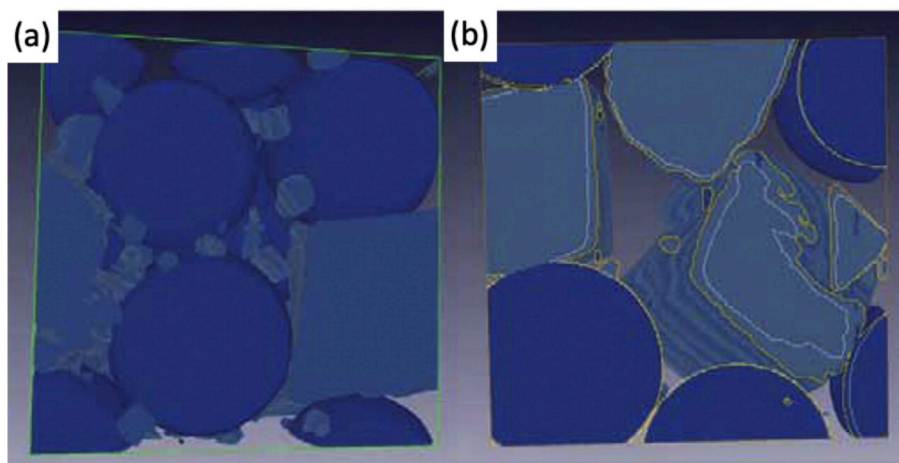
### 3.2. Pore size effect

Precipitation in porous media is strongly dependent on pore size. Field observation by Putnis and Mauthe [142] reported that halite crystallizes preferentially in larger pores. This observation is usually attributed to a thermodynamic hindrance to mineral nucleation in small pores, where mineral solubility is increased by confinement through curvature-induced interfacial energy and thus excess pressure within the crystal. Subsequently, the effective solubility deviates from the value in the bulk ( $K_{sp,bulk}$ ). The following equation has been proposed and used to account for this pore size control on solubility, aka Pore Controlled Solubility (PCS) model [37,38,167]:

$$K_{sp,d} = K_{sp,bulk} \exp(v_m \sigma \zeta / RT) \quad (17)$$

where  $K_{sp,d}$  is the effective solubility in a pore of size  $d$ , and  $v_m$  is the molar volume.  $\zeta$  is the crystal curvature, which is defined as the rate of change of volume with respect to surface area, and for a spherical crystal with radius of  $r_p$ ,  $\zeta = 2/r_p$ , and  $r_p$  is related to the pore size ( $d$ ) and contact angle ( $\theta$ ) by  $r_p = d/\cos\theta$ . This pore-size control on solubility is typically significant in pores below  $\sim 100$  nm (Fig. 3a).

Besides the thermodynamic effect, kinetics also play an important role. Nucleation is probabilistic, with the probability scaling with pore volume. Thus, induction time for nucleation is inversely proportional to pore volume, a hypothesis proposed by Prieto and co-workers based on counter-diffusion experiments in silica gels [140]. However, these experiments did not control individual pore sizes. More recently,



**Fig. 2.** Post-reaction computed tomography images of  $\text{CaCO}_3$  precipitation on glass beads (dark blue) and calcite spar (light blue), illustrating (a) discrete crystals on glass beads and (b) epitaxial growth on the calcite spar (the blue and yellow contours are the outlines of the grains at the initial and final stage). [115]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Poonoosamy et al. [131] used microfluidic setups to better constrain pore size and confirmed that nucleation probability scales with pore volume, showing that while the thermodynamic effect becomes important for pores below 100 nm, kinetics retardation may start in micro size pores (Fig. 3b). Yet, they also concluded that this relationship is not universal: pore surface functionalities, geometry, and roughness can reverse the trend, consistent with observations by Stack et al. [168].

In addition, the pore-size effect may be even more complex when nano-confinement is involved. In nanopores, fluid properties (e.g., water density and viscosity) may be altered [189]. Strengthening of the electrical double layer effect may result in stronger anion exclusion and different ion concentrations compared to the bulk or the larger pores, leading to different precipitation dynamics [151].

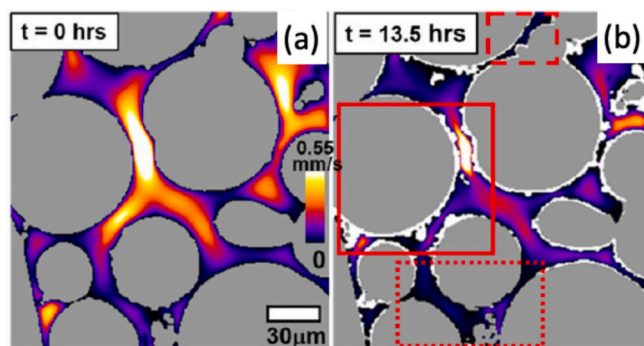
### 3.3. Local reactive transport conditions

Reaction rates observed in porous media are typically lower than those measured in the bulk [50,115,116], because of the variable local reaction rates that are dependent on local reactive transport conditions. Local reactive transport conditions are controlled by factors such as the structure of the fractured porous media and the boundary conditions (Sections 3.3.1 & 3.3.2), and the co-existence of precipitation and dissolution (Section 3.3.3).

#### 3.3.1. Flow of supersaturated solution

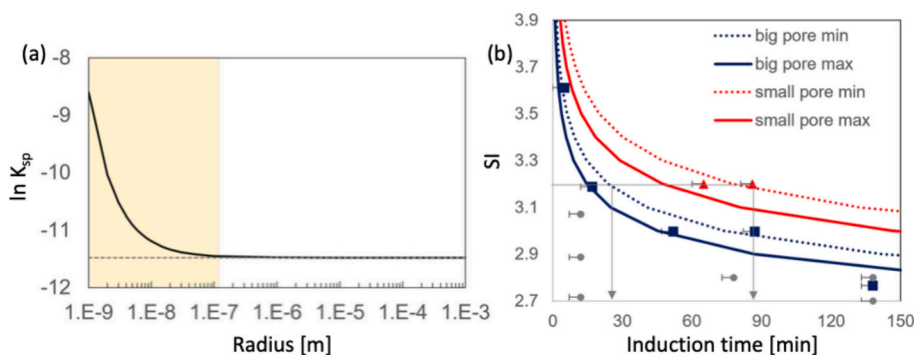
When supersaturated solutions are injected into fractured porous media, the observed precipitation rate and patterns are dependent on whether the precipitation process is surface reaction controlled or transport controlled. In surface reaction controlled regimes, typically associated with large Damkohler number – which is the ratio between the reaction rate and the advective ( $Da_I$ ) or diffusive ( $Da_{II}$ ) transport rate – overall precipitation rates are closer to the bulk and the precipitation is more spatially homogenous. In transport controlled regimes with small Damkohler numbers, local precipitation rate is highly correlated with the transport rate as it determines the supply of the reactants, and thus distribution of the precipitates within the porous media is heterogeneous. For example, in the barite precipitation experiment of Godinho et al. [49], CT images showed more precipitates in regions with higher velocities identified by the flow simulations (Fig. 4). But if the substrate in the high velocities are not favorable for precipitation, the extent of precipitation can decouple from local velocities and overall precipitation rate may decrease over time as the precipitation directs the flow away from the increasingly narrowed pathways [69,113].

The spatial heterogeneity in mineral precipitation is evident at the single grain scale. Fengchang Yang et al. [198] has shown the evolution



**Fig. 4.** Correlation between local velocity and barite (white) precipitation in  $\text{SiO}_2$  particle columns (gray), with warmer colour indicating larger velocities (a) before experiment and (b) after reacting 13.5 h at a flow rate of 60 ml/h and at  $SI = 2.1$ . The red solid-line box highlights the high velocity and heavily precipitated region, the dashed-line box shows regions with initial high velocity and thus densely nucleated area, and the dotted-line box is where the velocity is low and precipitation is rare [49]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of a precipitating grain switching from uniform coating to less uniform growth to dendrite formation (i.e., tree-like/branched precipitation patterns) (Fig. 5a), as the Damkohler number increases. This phenomenon has also been reported in [73,121,169,173] (Fig. 5b). Initial heterogeneity in the pore structures, e.g., presence of fractures, would result in even more significant spatial variations in the flow field, which will be amplified when the precipitation process is transport limited (low  $Da$ ) and especially at large Peclet numbers. The simulations of Tartakovsky, Meakin, Scheibe, and Wood [173] reported more localized precipitation at the fracture interface than into the matrix at increasing  $Da_{II}$ . Recent lattice Boltzmann model simulations observed four precipitation regimes in fractured porous media [4]. No precipitation is formed when the reaction rate is slower than the advection rate, and precipitation preferential forms at and clogs the inlet when the reaction rate is faster than the advection rate by far. When reaction rate and advection rate is somewhat comparable, precipitation blocks the mass transfer between fractures and porous media and seals the fractures gradually if diffusion is much slower than the advection, whereas if diffusion is fast, precipitation occurs and clogs the porous matrix. The microfluidic experiments of Q.-R. Jiang et al. [66] documented barite scaling on the fracture surface when the precipitation rate is fast compared to the flow and transport rate, whereas a barite precipitation band forms in the matrix when the precipitation rate is too slow compared to the flow rate in the fractures but comparable to the



**Fig. 3.** (a) Illustration of the change in effective solubility of barite with the pore radius, where the yellow shade highlights the PCS effect, and (b) the relationship between  $SI$  and induction time for barite precipitation in different pores, with the gray line and arrows highlighting the increase in induction time in smaller pores. The symbols show the experimental data in large pores (cylindrical pores of 20  $\mu\text{m}$  diameter and 1  $\mu\text{m}$  depth, blue square), small pores (cylindrical pores of 6  $\mu\text{m}$  diameter and 1  $\mu\text{m}$  depth, red triangle), and capillaries (squared capillaries with size of 1  $\mu\text{m}$ , gray circles), and the lines are calculations based on CNT where the min and max encompass values calculated from different  $\text{BaSO}_4$  monomer concentrations capturing the gradients present in the experimental pore network [131]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

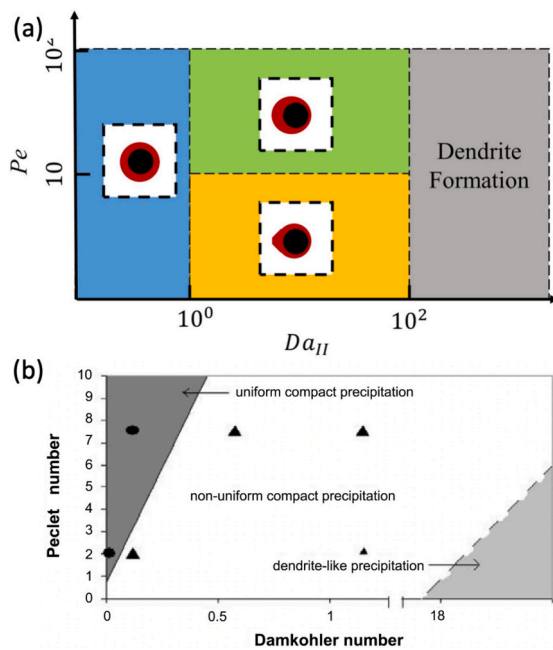


Fig. 5. Phase diagram showing the precipitation pattern at different Peclet number and Damkohler numbers in flowing supersaturated solutions in (a) a single grain system [198] and (b) a porous media [173].

diffusion rate in the matrix. In the first case, the scaling prevented mass transfer between fracture and porous matrix and thus precipitation within the matrix was minimized. If the precipitation reaction results in significant drop in the saturation state along the flow direction, the rate and amount of precipitation will decrease along the flow direction accordingly [115].

### 3.3.2. Mixing-induced precipitation

When supersaturation is reached by the mixing of two solutions, a characteristic precipitation band is formed, and the location of the precipitation will be predominantly determined by the mixing line [20,24,39,151,200,206]. Homogeneous nucleation in the solution and/or heterogenous nucleation on another substrate and/or growth on the same mineral phase occur in the mixing region of high SI [35,203]. The width of the precipitation band depends on local reactive transport conditions, increasing with decreasing  $Pe$  and/or  $Da$  [174], i.e., when diffusion plays an important role in determining local supersaturation. In both diffusion only and advection dominated mixing systems, a

thinner precipitation band was observed at a faster reaction rate due to e.g., higher supersaturation or specific surface area [35,206] (Fig. 6a-b). This is because the precipitates serve as a barrier that separates the two solutions and prevents further mixing. At a faster mineral precipitation rate, the suppression of mixing and thus precipitation kicks in earlier, typically resulting in a net effect of reduced precipitation.

Pore structure heterogeneity affects the mixing dynamics and thus the precipitation pattern [67]. In more heterogeneous porous matrix, pore scale simulations predicted a more tortuous mixing line and a wider precipitation band that crosses over more pores [35] (Fig. 6c). At the continuum scale, an inclusion of high permeability zone in an otherwise homogeneous porous media enhances dispersion and thus led to wider precipitation band in the inclusion, whereas a low permeability inclusion zone directs the mixing and precipitation to downstream of the inclusion [43].

The hydrodynamics or transport mechanisms that create the mixing condition (e.g., injection strategies, gravity and inertia) will also determine the characteristics of the precipitation band. Parallel or concentric injections of solutions result in precipitation band that is parallel to the flow direction [15,206], whereas counter diffusion causes precipitation band to form perpendicular to the transport direction [20]. The relative magnitude of the two injection fluids also affects the location and extent of precipitation, with a wider precipitation band being created when the injection rates are unequal and dispersion is enhanced [15]. The contribution of gravity may be non-negligible in wide open fractures and at high SI when homogeneous nucleation is dominant, as it influences the transport of the precipitated (nano)particles and that density changes in the solution due to the precipitation reaction can cause convection and different mixing patterns [196]. The study observed that in vertical fractures, gravity driven flow can confine the mixing around the injected stream of the carbonate rich solution and resulting in precipitate buildup around it, whereas in horizontal fractures, the precipitates will settle onto the bottom and accumulate to block the fracture. At increased flow velocities when inertia becomes important, the mixing zone is featured by vortices that enlarge the precipitation zone rather than dense streamlines in a narrow region [200]. Unequal mixing was also shown to reshape the fluid inertial effect, leading to enhanced mixing and precipitation in the downstream. Other strategies that enhance mixing e.g., via oscillations and pulsations [109] are expected to increase precipitation, provided that the precipitates do not block the mixing.

### 3.3.3. Coupled precipitation and dissolution

In cases when precipitation is coupled with mineral dissolution or redox reactions, the dependence of precipitation rate and pattern on local reactive transport conditions is more complex. Fast flow is

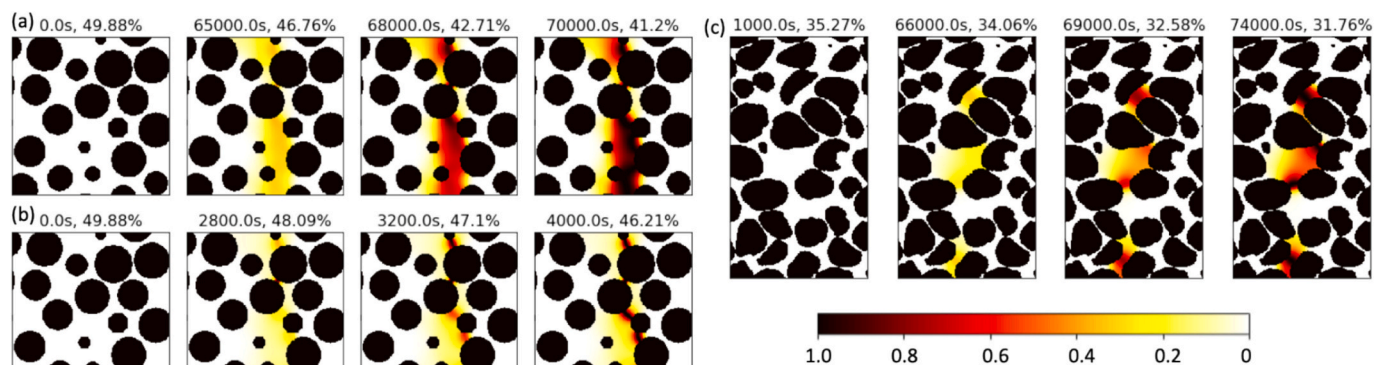


Fig. 6. Snapshots of mixing-induced precipitation in a counter-diffusion setup, in (a-b) a spherical grain geometry and (c) a quartz sand pack, with a low specific surface area and thus low precipitation rate (a,c) and a 100 times higher specific surface area (b). The colorbar shows the volume fraction of the precipitates. Note that due to the difference in reaction rate and geometry, each case reached the clogging point at different time steps, and the snapshots shown were chosen accordingly. [35].

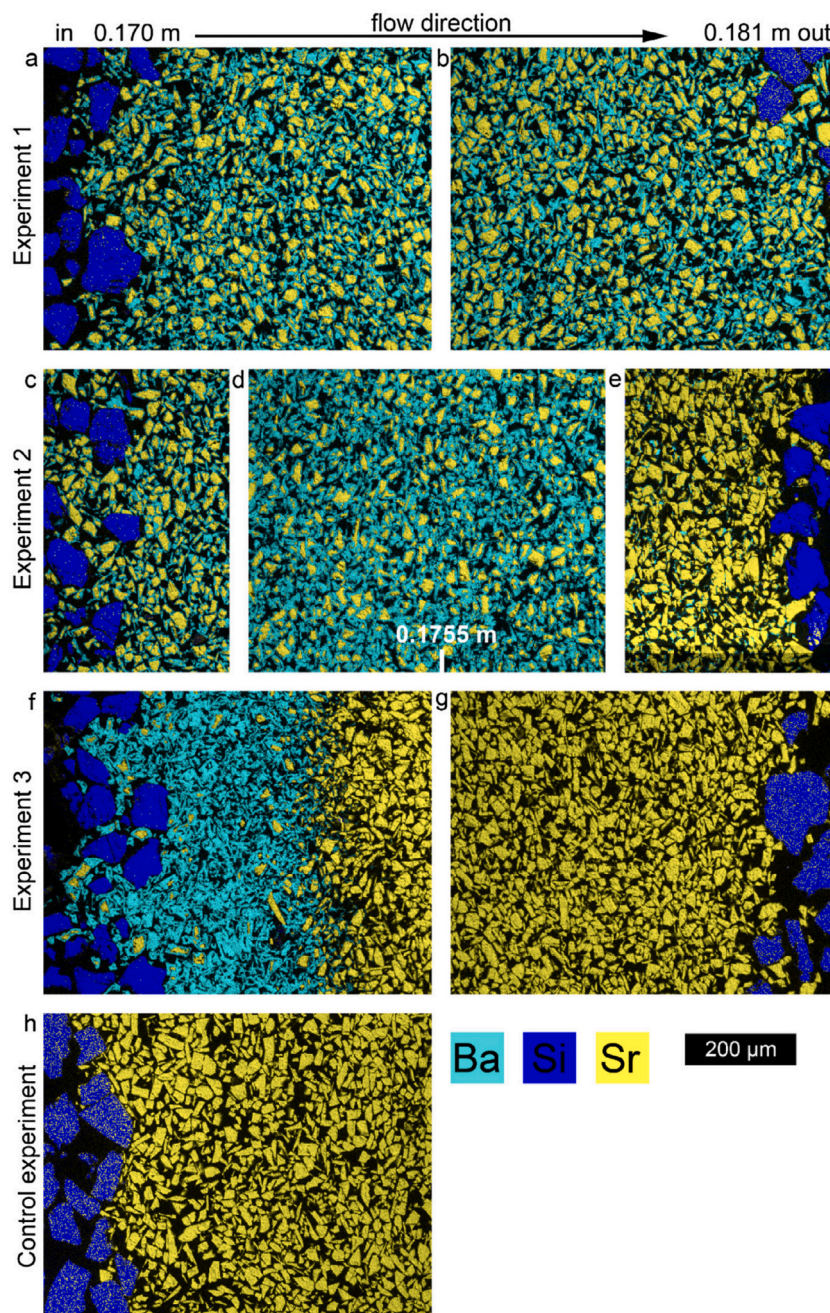


necessary to achieve high dissolution rate, but transport limited regions are key to retain the solutes to reach supersaturation. In these cases, preferential precipitation is typically observed in diffusion-dominated micro-pores or pore throats [46,94,184] or in the matrix away from fractures [190]. The core-flooding experiments of Xiong et al. [194] provide an example of the contrasting precipitation behaviors, observing barite deposition from the initially over-saturated influent on the fracture surfaces, and Fe-hydroxide precipitation in the shale matrix following pH increase from calcite dissolution and oxidation of Fe(II)-bearing minerals.

Overall, the extent of precipitation depends on the relative dissolution rate and precipitation rate and the flow velocity [162]. When the dissolution rate is faster than the precipitation rate, the dissolution reaction dictates the alteration of the fractured porous media [46,94],

whereas the precipitation reaction is dominant and shapes the porous structure when the precipitation rate is faster [162]. A competition between the dissolution and precipitation reaction may arise when the two reaction rates are comparable [162]. The amount of precipitates may increase with flow rate when the supply of the reactants for the precipitation reaction is controlled by the dissolution reaction. But when the flow rate increases to a point that local fluid renewal is fast enough to prevent the solution from reaching supersaturation, precipitates can no longer be produced [46].

One special case is interface coupled dissolution and precipitation (ICDP), which occurs when precipitation is fast compared to transport dominated by the boundary layer at the interface [143,144]. As a result, the secondary precipitating mineral tracks the morphology of the parent mineral, and the level of conformity depends on the molar volume



**Fig. 7.** Post core-flooding experiment EDS maps of the upstream (a, c, f, and h), middle section (d), and downstream interfaces (b, e, and g) of the dissolving celestite (yellow) and precipitating barite (cyan) with influent Ba concentration and thus SI decreasing from experiment 1 to 3. [128]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



differences and the microporosity in the secondary mineral phase. The microscopic alteration is also coupled with the macroscopic reactive transport conditions. For example, experimental and numerical simulations have shown different macroscopic patterns along the flow direction for barite precipitation following celestite dissolution at different influent  $\text{Ba}^{2+}$  concentrations and thus SI values [33,128] (Fig. 7). At the high influent concentration and high SI, the precipitation is observed to uniformly coat the entire core, whereas at the low influent concentrations, the precipitation is highly localized at the inlet, and at an intermediate influent concentration, the precipitation is maximized at the middle section instead. This is because at the high SI local precipitation rate is fast, it rapidly coats the parent mineral and prevents further dissolution and thus the release of sulfate ions. Consequently, reactants migrate downstream where precipitations continues.

#### 4. Impacts of mineral precipitation on physical-chemical properties of porous media

Mineral precipitation in fractured porous media results in morphological and geochemical alteration, which manifests as evolving bulk physical-chemical properties. An important goal of improving our understanding of the mineral precipitation process is to provide insights and develop quantitative relationships for predicting precipitation driven alteration in fractured porous media.

#### 4.1. Permeability

##### 4.1.1. Commonly used porosity-permeability relationships

Permeability change of an evolving porous media is typically described by a porosity ( $\phi$ )-permeability ( $k$ ) relationship, such as the Kozeny-Carman relationship, different modifications of the KC relationship, exponential relations, and power-law relations [59]. The typical forms of the KC relationship includes:

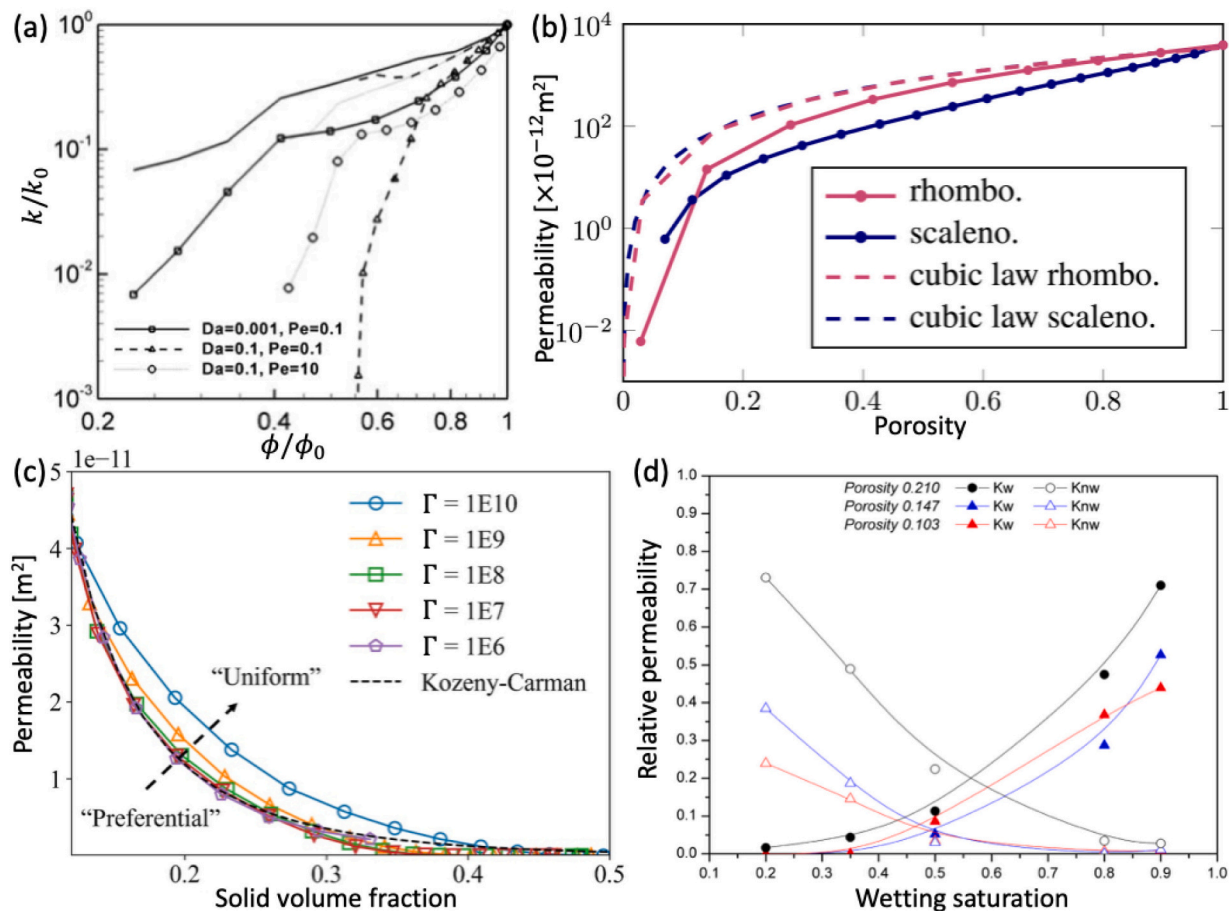
$$k = k_0 \left( \frac{1 - \phi_0}{1 - \phi} \right)^2 \left( \frac{\phi}{\phi_0} \right)^3 \quad (18a)$$

$$k = k_0 \frac{\phi^3 S_0}{\phi_0^3 S} \quad (18b)$$

$$k = \frac{\phi^3}{\tau(1 - \phi)^2 s^2} \quad (18c)$$

where  $\tau$  is the tortuosity and  $s$  is the specific surface area.

The KC relationship is the most widely used in continuum scale reactive transport modeling [191], including systems involving mineral precipitation [18,130,133]. It has been shown to properly capture precipitation-driven porosity-permeability changes in porous media when the underlying assumptions (e.g., granular pack structure) hold throughout the geochemical reactions [9,51,171], and even for simple fractures [87]. However, the KC relationship may fail to capture



**Fig. 8.** (a) Porosity-permeability relationships predicted by the KC relationship (Table 2) (solid lines) and from 3D pore-scale simulations of an open fracture with an initial sinusoidal aperture (symbolled lines) at different reactive transport conditions [87], (b) porosity-permeability relationships predicted by the cubic (Table 2) (dashed lines) and from 3D pore-scale simulations of precipitation of calcite of different habits in open fractures [166], (c) permeability evolution following barite precipitation with different nucleation rates [197], and (d) evaluations of relative permeability based on pore structures undergoing different levels of clogging following precipitation predicted by reactive transport modeling [65].

permeability changes caused by mineral precipitation when the resulting structures deviate from the underlying assumption [51], and when clogging happens [87,128] (Fig. 8a).

Instead, the relationship (Eq. (19)) proposed by Verma and Pruess [180] that introduced a critical porosity ( $\phi_c$ ), i.e., the porosity at which the porous media becomes effectively impermeable, has been shown to provide a better fit to experimental data (Table 2).

$$k = k_0 \left( \frac{\phi - \phi_c}{\phi_0 - \phi_c} \right)^n \quad (19)$$

$\phi_c$  provides a degree of freedom that can be used to account for the clogging behavior that may arise from mineral precipitation [9,171]. The exponent  $n$ , similarly to the power law relationship, can be adjusted to capture characteristics of structural alteration, with larger exponent corresponding to more spatially heterogeneous precipitation as expected under transport limited conditions [110,188].

**Table 2**

Porosity-permeability relationship used for evolving porous media driven by mineral precipitation.

| $k - \phi$ relationship                                                                                                                                                                                                    | Coefficients                                                                                                                                                                                                                                                                                                                           | System description                                                                                | Reference |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------|
| Kozeny-Carman relationship<br>$k = \frac{\phi^3}{\tau(1 - \phi)^2 s^2}$                                                                                                                                                    | $\tau = \tau_0 \left( \frac{\phi}{\phi_0} \right)$<br>$s$ : specific surface area explicitly evaluated                                                                                                                                                                                                                                 | 3D level-set simulations of fractures with an initial sinusoidal aperture                         | [87]      |
| Verma-Pruess equation<br>$k = k_0 \left( \frac{\phi - \phi_c}{\phi_0 - \phi_c} \right)^n$                                                                                                                                  | $\phi_c = 0.86\phi_0$<br>$n = 5$                                                                                                                                                                                                                                                                                                       | Celestite grain dissolution and barite precipitation experiments                                  | [128]     |
| Dual-exponential function<br>$k = ae^{b\phi} + ce^{d\phi}$                                                                                                                                                                 | $a = 1.64 \times 10^{-14}$<br>$b = 18.32$<br>$c = 7.86 \times 10^{-29}$<br>$d = 250.6$                                                                                                                                                                                                                                                 | Pore-network modeling, assumed precipitation in voids >30 $\mu\text{m}$                           | [9]       |
| Kozeny-Carman relationship<br>$k = \frac{\phi^3}{\tau(1 - \phi)^2 s^2}$                                                                                                                                                    | $\tau = 2.5$<br>$s = 53691.3$                                                                                                                                                                                                                                                                                                          | Pore-network modeling, uniform reaction                                                           | [9,171]   |
| Verma-Pruess equation<br>$k = k_0 \left( \frac{\phi - \phi_c}{\phi_0 - \phi_c} \right)^n$                                                                                                                                  | $\phi_c = 0.3\phi_0$<br>$n = 1, 2$                                                                                                                                                                                                                                                                                                     | Pore-network modeling (sandstone), uniform reaction                                               | [9]       |
| Verma-Pruess equation<br>$k = k_0 \left( \frac{\phi - \phi_c}{\phi_0 - \phi_c} \right)^n$                                                                                                                                  | $\phi_c = 0.3\phi_0$<br>$n = 0.9$                                                                                                                                                                                                                                                                                                      | Pore-network modeling, artificial size-dependent precipitation, large pores and pore throats      | [171]     |
| Modified Verma-Pruess equation<br>$k = k_0 \left( \frac{\alpha\phi_0}{(1 + \alpha)\phi_0 - \phi} \right)^n$                                                                                                                | $\alpha = 0.02, n = 0.5$ ; $\alpha = 0.04, n = 0.5$ ;<br>$\alpha = 0.1, n = 0.5$ ; $\alpha = 0.12, n = 0.6$ ;                                                                                                                                                                                                                          | Pore-network modeling, assumed size-dependent precipitation, small pores and pore throats         | [171]     |
| $k^{t+1} = k^t \left( \frac{\tau^t}{\tau^{t+1}} \right)^2 \left( \frac{\phi^{t+1}}{\phi^t} \right)^3 \left( \frac{1 - \phi^t}{1 - \phi^{t+1}} \right)^2$                                                                   | tortuosity $\tau = \phi^{-3/4}$                                                                                                                                                                                                                                                                                                        | Continuum scale modeling of Fe precipitation                                                      | [18]      |
| Power law relationship<br>$k = k_0 a \left( \frac{\phi}{\phi_0} \right)^n$                                                                                                                                                 | $n$ highly variable                                                                                                                                                                                                                                                                                                                    | LBM modeling with considerations of stochastic nucleation                                         | [97]      |
| Modified Kozeny-Carman relationship<br>$k = k_0 \left( (1 - \alpha) \left( \frac{\phi/\phi_0}{s/s_0} \right)^3 + \alpha \right)$<br>$\alpha = \exp \left( a + \frac{b}{\ln(h)} \right)$<br>$h' = (h - 1) + \sqrt{1 - w^2}$ | $a = 0.406$<br>$b = 1.906$<br>$w$ : crystal width<br>$h$ : center height of the crystal<br>$s$ : surface area of the circular tube                                                                                                                                                                                                     | CFD modeling considering explicit crystal geometry in circular tubes                              | [48]      |
| The cubic law<br>$k = k_0 \left( \frac{\phi}{\phi_0} \right)^3$                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                        | 3D phase-field simulations of precipitation in open fractures with calcite of rhombohedral shapes | [166]     |
| $k = k_0 \tau^{1/2} \left( (\delta - 1) \left( 1 - \left( 1 - S_w^{1/m} \right)^m \right) + 1 \right)^2$                                                                                                                   | $S_w$ effective water saturation<br>$\beta = (\theta_p - \theta_{salt}) / \theta_p$<br>$\delta = 4.5\beta$<br>$\tau = 1 - S_p + \delta^2 S_p$<br>$\theta_p, S_p$ water content and water saturation when precipitation occurs,<br>$\theta_p$ : content of precipitated salt<br>$m$ : empirical exponent of the van Genuchten relation. | Salt precipitation in porous media                                                                | [91]      |
| $k = k_0 \tau^{1/2} \left( (\delta\beta - 1) \left( 1 - \left( 1 - S_w^{1/m} \right)^m \right) + 1 \right)^2$                                                                                                              |                                                                                                                                                                                                                                                                                                                                        | Salt precipitation in porous media                                                                | [207]     |
| $k = k_0 \left( \frac{\phi - \phi_c}{\phi_0 - \phi_c} \right)^a \left( \frac{1 - \phi_0 - \phi_c}{1 - \phi - \phi_c} \right)^b$                                                                                            | $\phi_c = 0.08$<br>$a = 2.5$<br>$b = 1.5$                                                                                                                                                                                                                                                                                              | Salt precipitation in sandstone core                                                              | [93]      |

To capture permeability evolution in cases involving complex precipitation patterns and structural alteration, empirical relationships composed of two exponential functions [9] or step functions [171] were proposed based on pore network modeling results (Table 2). In the modified step functions, while the Verma-Pruess relationship was retained for the precipitation in pore networks with large pores and pore throats, an additional fitting parameter was introduced for cases with small pores and pore throats (Table 2).

#### 4.1.2. Permeability evolution caused by dissolution and precipitation

In cases when dissolution and precipitation are both involved, permeability evolution possesses new features.

Hysteresis in the porosity-permeability relationship has been reported for cyclic dissolution and precipitation reactions, i.e., multiple porosity-permeability trajectories were recorded throughout the experiments [188]. This hysteresis suggests that at any given time point, the flow patterns are dependent on the porous structures that are shaped by

the reaction history, which in turn determine the location and extent of reactions in the next time step. This history-dependence on reaction path generates different trajectories in pore-structure and permeability evolution, and hence the hysteresis. The underlying mechanism is consistent with the need of constructing a two-step function for describing the porosity-permeability relationship for the precipitation-driven versus dissolution-driven alteration of the same pore networks [171].

In co-dissolution and precipitation systems, oscillations in the evolution of both porosity and permeability were observed as the dissolution and precipitation reaction competes with each other [16,162]. The onset of the oscillation is featured by the initial dissolution (i.e., permeability increase) that releases the solutes necessary for the precipitation reaction, and the subsequent precipitation (i.e., permeability decrease) when supersaturation condition is reached from the accumulation of the solutes. Afterwards, the oscillatory behaviors continue because of the spatially variable interplay between flow, diffusion and reactions. While local dissolution opens up pores and flow paths, local precipitation leads to narrowing of flow path and higher local flow resistance. The overall response depends on whether the dissolution or precipitation dominates. The pore-network modeling of Budek and Szymczak [16] also illustrated that as precipitation tends to form at the reaction front where the solutes are abundant, it blocks the penetrating path; while the increase in permeability in each oscillatory cycle likely corresponds to an event of percolation, i.e., the formation of a preferential flow path caused by mineral dissolution [16].

While the observation of hysteresis was based on dissolution and precipitation that are decoupled [188], it is reasonable to submit that the porosity-permeability relationship for a co-dissolution and precipitation system will show similar characteristic, i.e., the porosity-permeability relationship for each of the oscillatory cycle will be distinctive, since the oscillatory behaviors are the result of local interplays between transport and mineral reactions.

#### 4.1.3. Permeability evolution with nucleation and mineral transformation

The discussions above did not explicitly consider the nucleation process that is also key in controlling the precipitation process and thus the evolution of permeability. In the experiments of Noiri et al. [116], precipitation of calcium carbonate resulted in more uniform coating on the calcite grains, and roughening on the glass beads and aragonite grains. Permeability was evaluated using flow simulations based on porous structures reconstructed from the post-reaction CT images and on porous media generated based on different nucleation and growth scenarios. It was shown that the decrease in permeability given a comparable porosity reduction is more significant in cases with non-uniform nucleation and/or growth, which is the case when the substrates are glass beads or aragonites. Ghezzehei [48] investigated the permeability in a circular tube with discrete crystals distributed on the wall, and developed a modified porosity-permeability relationship (Table 2). The numerical investigations of precipitation in open fractures by Späth and Nestler [166] further highlighted that the porosity-permeability relationship also depends on the habit of the crystals. The commonly used cubic law for fractures is applicable when calcite crystal is rhombohedral, whereas it deviates from the simulation results notably when calcite crystals have a scalenohedral shape (Fig. 8b).

According to the CNT, nucleation is caused by perturbations in the thermodynamically equilibrium solution, and is thus intrinsically stochastic. This further complicates the impacts of mineral precipitation on the evolution of pore structures and their permeabilities. The modeling study of Yang et al. [197] highlighted that consideration of the probabilistic nucleation would lead to different permeability evolution compared to the deterministic approach. With larger pre-exponential term and thus more uniform nucleation, the resulted porosity-permeability relationship deviates more from the KC relationship (Fig. 8c). Pore-scale lattice Boltzmann simulations have reported that different realizations of stochastic nucleation of crystallites in porous media can result in different exponents for a power law porosity-

permeability relationship [97]. Values between 2 and 18 were reported for homogeneous porous media, and a wider range of 2 to >30 were fitted for heterogeneous porous media.

Recent study has shown that the transformation of the precipitating phase can also affect the evolution of the permeability pathway. Xu and Balhoff [195] used microfluidic experiments and lattice Boltzmann simulations to investigate mixing induced calcium carbonate precipitation. At a relatively high SI (>6), amorphous calcium carbonate (ACC) formed initially, generating a thin film along the mixing line (i.e., localized precipitation). The ACC film may rupture following the transformation of ACC to the crystalline phase, which resumes the mixing and continued precipitation. This results in deformed and/or secondary precipitation films/bands and the spread-out of the crystalline phase (i.e., intermittent precipitation) or the ACC phase (i.e., homogeneous precipitation). With localized precipitation, the power law relationship with a single exponent can be fitted for the pressure measurement derived permeability and porosity relationship; whereas the exponent was observed to change significantly during the experiments for intermittent precipitation and homogeneous precipitation, with a much larger exponent at the early stage.

#### 4.1.4. Multiphase flow effect

When coupled with multiphase/two phase flow, mineral precipitation and the resulting permeability change become more complex. It has been documented that following the structural changes due to mineral precipitation, relative permeability of the non-wetting phase is affected more than the wetting phase [65] (Fig. 8). This is a result of disproportionate alteration in pores and pore throats of different sizes, as the non-wetting phase tend to occupy large pores, where the wetting phase populates the smaller pores and pore throats.

The complexity of the multiphase system also arises from the fact that the multiphase flow determines the distribution of the aqueous phase and thus where the precipitation reaction can occur at the fluid-solid interface, as well as controls the gas-fluid interfaces, which serve as another interface for mineral precipitation [56,105,117,187]. In salt precipitation microfluidic experiments, the VP porosity and permeability relationship with a carefully chosen threshold value was able to capture the general trend, but the experimental data are more scattered, suggesting the presence of other underlying mechanisms [56]. For example, at the liquid-gas interface, polycrystalline aggregates dominate rather than large well-developed crystals in the bulk, partly due to the capillary effects and thus increased interface for continued evaporation and salt precipitation caused by the micro-/nano-pores in the precipitates [105]. Lopez et al. [93] used a modified KC relationship to model the core flooding experiments in which salt precipitates in sandstone samples following dry gas injection (Table 2). Other permeability relationships were also developed for modeling systems with salt precipitation, by introducing the amount of salt precipitation and other parameters associated with salt precipitation in the van Genuchten relationship [91,207].

#### 4.1.5. Effect of heterogeneity

The porosity-permeability relationships do not explicitly account for the effects of structural and mineralogical heterogeneity. While the effect of precipitation in structurally heterogeneous fractures and porous media can be captured by parameter fitting as discussed in 4.1.1, the capability of existing porosity-permeability relationships in capturing the effect of mineralogical heterogeneity haven't been fully assessed. Whereas the simulations of Jones and Detwiler [70] have already showed that heterogeneity in the mineral distribution shortens the sealing time of the fracture compared to a mineralogically homogeneous fracture; and this is especially evident under lower  $Da$  conditions.

One other aspect that was typically missed by the porosity-permeability relationship is the anisotropy of permeability, which may be enhanced by precipitation and initial heterogeneity [97,113]. The experimental work of Cil et al. [24] has documented a larger



permeability decrease in the direction perpendicular to the flow or to the mixing line following mixing induced precipitation; whereas injection of oversaturated solution caused more permeability reduction along the flow direction.

#### 4.2. Impacts of mineral precipitation on diffusivity

Effective diffusivity ( $D_{eff}$ ) is a parameter that accounts for the impact of tortuosity on solute diffusion in porous media. It is typically evaluated from porosity using Archie's law [6,161].

$$D_{eff} = \phi^m D_m \quad (20a)$$

$$D_{eff} = \phi \tau D_m \quad (20b)$$

where  $D_m$  is the molecular diffusion coefficient, and  $m$  is the cementation exponent, which typically ranges between 1.0 and 2.0 and is dependent on the type of rocks and the diagenesis history [205].

The porosity-diffusivity relationship is a key constitutive relation used in reactive transport modeling of evolving porous media [191]. Similar to permeability, effective diffusivity changes in response to pore structure alteration caused by mineral precipitation. Typically, when the heterogeneity of pore structure increases due to mineral precipitation, the cementation exponent becomes larger [110].

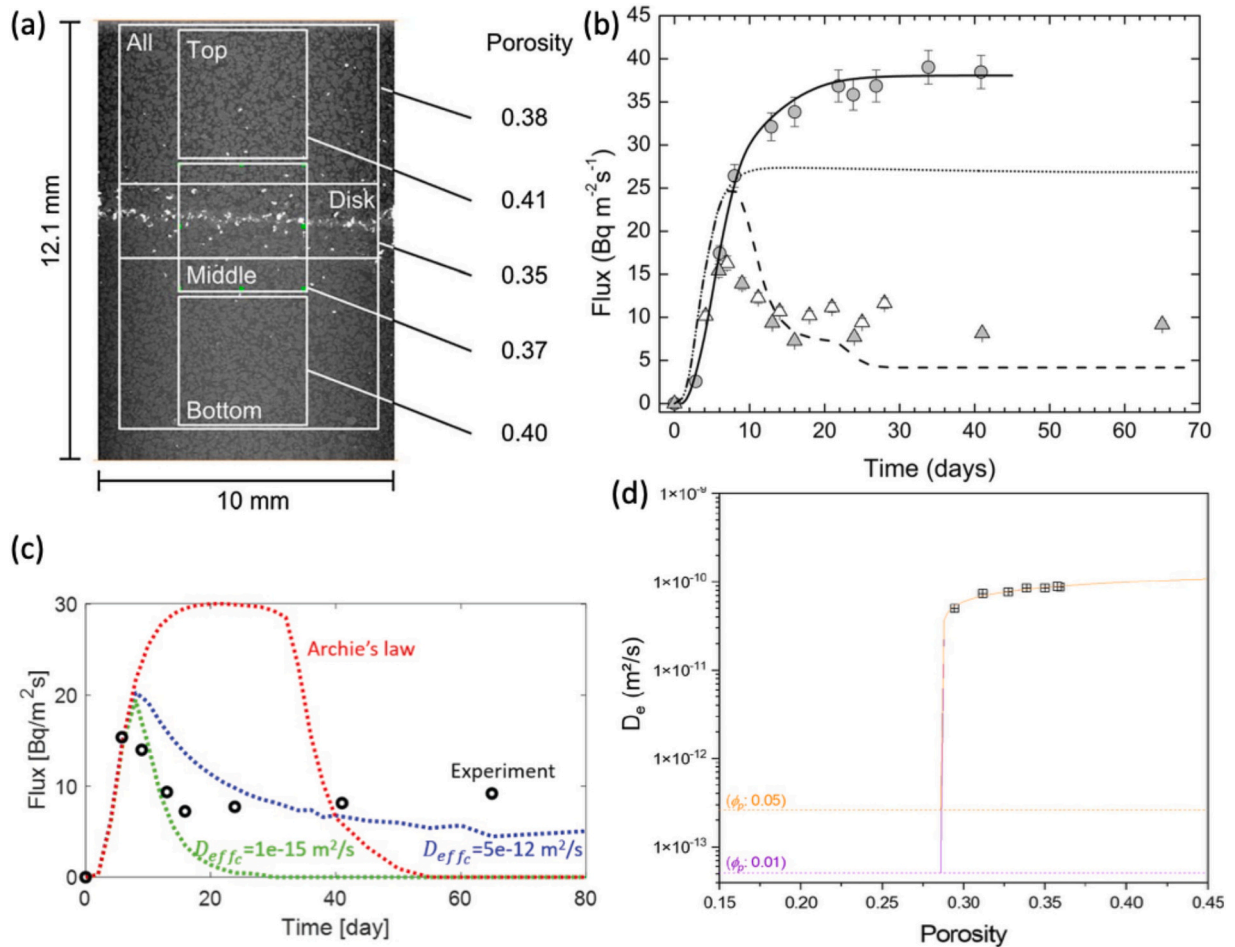
However, the conventional form of Archie's law may fail to capture

precipitation-driven diffusivity evolution. For example, in the counter-diffusion experiments of Chagneau et al. [20], celestite precipitation in sand columns was observed resulting in a small porosity reduction where the two solutions mix (Fig. 9a). But the corresponding diffusivity change as indicated by the diffusion of tritiated water (HTO) is much smaller than what was predicted by the continuum-scale model using Archie's law. As a result, the continuum reactive transport model that uses the traditional Archie's law fails to reproduce bulk behaviors of mineral precipitation and diffusivity evolution regardless of the cementation exponent used (Fig. 9b). The simulation that reproduces tracer flux predicted incorrect amount of precipitation.

The pore-scale modeling investigation of Deng et al. [35] was able to reproduce the sharp decrease in diffusivity following the limited porosity reduction as observed in the experiments. Conceptualizing precipitation as either nucleation or surface growth in both idealized spherical packs and realistic quartz sand packs generated porosity-diffusivity trajectories that are qualitatively consistent. A modified version of the Archie's law was proposed, which introduced a critical porosity ( $\phi_c$ ) at which the porous media is effectively clogged.

$$D_{eff} = D_{eff,c} + a D_m (\phi - \phi_c)^n \quad (21)$$

where  $a$  and  $n$  are empirical parameters, and  $D_{eff,c}$  is the effective diffusivity at the critical porosity, and is small but nonzero. Even though at effective clogging, solute transport is too slow to sustain continued



**Fig. 9.** Counter diffusion experiments of [20]: (a) CT image showing the celestite precipitation band (white in the middle), and (b) HTO flux from the experiments without (circles) and with (triangles) precipitation and from the continuum scale models disabling (solid line) and enabling precipitation with a minimum porosity of 0.01 (dotted line) and 0.3 (dashed line). (c) continuum scale modeled flux of the same experiment using the modified archie's law [35], and (d) experimental data of effective diffusivity in celestite and the fittings using Eq. (21) assuming different microporosity in celestite and the corresponding critical effective diffusivity is  $5.1 \times 10^{-14}$  and  $2.6 \times 10^{-13}$  m²/s [92].

precipitation at the observational time scale, it is not completely turned off due to the presence of porosity in the precipitates. The simulations also showed that compared to homogeneous nucleation, surface growth resulted in more effective clogging of the porous media, and hence a larger exponent. The exponent and pre-exponential parameter were observed to vary significantly with the initial geometry as well, highlighting the impacts of initial pore structures on mineral precipitation and the evolution of porous media transport properties (Fig. 9c).

This extended version of Archie's law has been corroborated by microfluidic experimental data [92] for a geochemical system similar to that of Chagneau et al. [20]. The experimental study also used Raman Spectra to detect the breakthrough of deuterium oxide through the neoformed precipitates. Considering a microporosity of 0.01 and 0.05 in the precipitates, the inferred critical effective diffusivity are  $5.1 \times 10^{-14}$  and  $2.6 \times 10^{-13}$  m<sup>2</sup>/s, respectively (Fig. 9d), which are in general agreement with the value reported in H. Deng et al. [35].

The microfluidic experiments of Poonoosamy et al. [129] showed that the extended Archie's law is also applicable for predicting diffusivity changes caused by coupled mineral dissolution of celestine and precipitation of strontianite before clogging. Because  $D_{eff,c}$  is far smaller than the second term before clogging, Eq. (21) can be re-written into a similar form as the VP porosity-permeability relationship (Eq. (19)). The results also highlighted that while the total porosity increased as a result of a smaller molar volume of the precipitating phase than the dissolving mineral phase, accessible porosity decreased because of the porous nature of the precipitates, resulting a decrease in the effective diffusivity. The same formulation has been implemented in the coupled reactive transport code HYTEC [176], and used to simulate barite and gypsum precipitation in chalks [150].

$$D_{eff}^t = D_{eff}^0 \left( \frac{\phi^t - \phi_c}{\phi^0 - \phi_c} \right)^n \quad (22)$$

One important aspect of the discussions on effective diffusivity is the diffusion of charged ions. In the counter diffusion experiments of A. Rajyaguru et al. [151], clogging resulted in a stronger impediment on the transport of charged ion (i.e., Cl<sup>-</sup>) than on the neutral molecules. This was attributed to the increased double electric layer effect associated with the small pores created by mineral precipitation (Fig. 10). Within

nanopores, electric double layers at mineral surfaces significantly influence cation and anion transport [201]. The mineral's zeta potential, which is dependent on the surrounding aqueous composition, can vary and generate local electrostatic gradients, further modulating ion migration through the nanoporous medium. This ion exclusion effect is evident in clay minerals that are abundant with nano-pores [21].

For unsaturated porous media, the impacts of multiphase flow on effective diffusivity are accounted for via introducing the water saturation into tortuosity [104]:

$$\tau_a = S_a^{7/3} \phi^{1/3} \quad (23)$$

It can be postulated that this correction alone will not be able to account for the pore clogging effects that may arise from mineral precipitation. However, to the best of our knowledge, no investigations have been conducted to systematically evaluate the impacts of mineral precipitation on diffusivity in unsaturated porous media.

#### 4.3. Impacts of mineral precipitation on reactivity

##### 4.3.1. Commonly used surface area and porosity relationships

Surface area ( $A$ ) is a key parameter for calculating mineral reaction rate. For simple geometries, specific surface area can be readily formulated based on the grain/pore size and thus the porosity. Assuming spherical grains gives the formulation in Eq. (24a), which results in a negative correlation between surface area and porosity [89]; whereas assuming spherical pores results in a positive correlation (Eq. (24b)) [79].

$$A = A_0 \left( \frac{1 - \phi}{1 - \phi_0} \right)^{2/3} \quad (24a)$$

$$A = A_0 \left( \frac{\phi}{\phi_0} \right)^{2/3} \quad (24b)$$

As such, in continuum-scale reactive transport models, mineral surface area is typically updated following a function of porosity and mineral abundances, as shown in Eq. (25a) for the primary minerals and Eq. (25b) for the secondary minerals [170], and in Eq. (25c) for fractures.

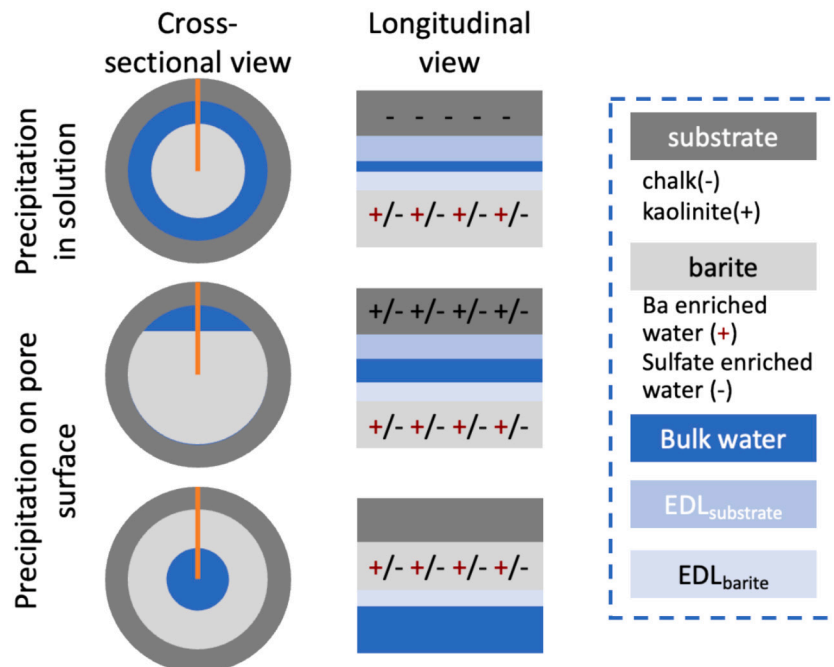


Fig. 10. Schematic view of the impacts of barite precipitation (light-gray) on pore geometry, and the resulting effect on electric double layer. Adapted from [149].

$$A_m = A_{m,0} \begin{cases} \left( \left( \frac{\phi}{\phi_0} \right) \left( \frac{\phi_m}{\phi_{m,0}} \right) \right)^{2/3} & \text{dissolution} \\ \left( \frac{\phi}{\phi_0} \right)^{2/3} & \text{precipitation} \end{cases} \quad (25a)$$

$$A_m = A_{m,0} \begin{cases} \left( \left( \frac{\phi}{\phi_0} \right) \phi_m \right)^{2/3} & \text{dissolution} \\ \left( \frac{\phi}{\phi_0} \right)^{2/3} & \text{precipitation} \end{cases} \quad (25b)$$

$$A_m = A_{m,0} \left( \frac{\phi}{\phi_0} \right) \quad (25c)$$

where  $\phi_m$  is the volume fraction of mineral  $m$ , and the subscript 0 denotes the initial porosity and surface area.

These relationships are based on the assumptions of simple geometries and that mineral-specific surface area is only related to its abundance. However, they may not be true in realistic and evolving porous media. The image analyses of sandstone and shaly sandstones by Peters [124] highlighted that the abundance of the mineral does not necessarily reflect the accessibility of the mineral. Clay minerals, which are common secondary minerals during geochemical alteration and have large specific surface area, tend to coat mineral grains and thus lower the surface areas of mineral phases such as quartz. Qin et al. [147] characterized accessible surface area of rock samples before and after being exposed to acidic solutions based on 2D SEM images and 3D CT scans. For both dissolving and precipitating minerals, the specific surface area estimated based on Eq. (25) are significantly different from image-based accessible surface area. In some cases, changes in the opposite direction were predicted.

In the column experiments of Noiriél et al. [115], roughness and surface area were observed to increase as a result of precipitation according to both BET analyses and image analyses (which is smaller due to its resolution 4.46  $\mu\text{m}$ ), regardless of the substrates of the columns. The experimental observations were reproduced by continuum scale reactive transport modeling using linear and non-linear rate laws, provided that the surface area is increased from 0.012  $\text{m}^2/\text{g}$  for the initial porous media to 0.21  $\text{m}^2/\text{g}$  for the neoformed phase. The later experimental study reported more detailed data that highlight the effects of the substrate [116]. In the mixed calcite spar and glass beads column, for a subvolume that experienced a porosity reduction from 36.3 % to 32.3 %, about 82 % of the precipitates were found on calcite spars, covering 100 % of the initial surface area and resulting in a 7 % increase in the calcite surface area. In the column of mixed aragonite ooids and glass beads, the porosity decreased from 36.3 % to 32.8 %, and 78 % of the precipitation occurred on 92 % of the aragonite ooids surface, leading to an increase of 41 % in the surface area of the aragonite ooids at the final stage. These changes in surface area cannot be captured by the constitute relations, all of which predicted a decrease, except for the relation used for dissolution of the primary mineral (Eq. (25a)), which predicted a  $\sim 6$  % increase in the surface area of calcite spar and  $\sim 5$  % increase in the surface area of aragonite. The prediction for the aragonite case is far lower compared to the experimental observation, primarily because the precipitating crystals have caused roughening on the aragonite surface and thus disproportional increase in the surface area.

#### 4.3.2. Surface passivation

The impacts of mineral precipitation on reactivity are even more complicated in cases of interface coupled dissolution and precipitation (ICDP). As the secondary mineral precipitates on the dissolving parent mineral phase, it covers the surface of the parent mineral phase and passivates the dissolution reaction. In the modeling the ICDP process, the following strategy has been developed:

$$A_p = V_p \times s_p \quad (26a)$$

$$A_d = s_d (V_{d,0} - K_p V_p^p) \quad (26b)$$

where the subscripts  $d, p$  denote the dissolving parent mineral phase and the precipitating mineral phase, respectively,  $s$  is the specific surface area of a given mineral phase, and  $V$  is the volume of the mineral,  $K_p$  is the proportionality factor and is typically a fitting parameter as  $p$ .

Because this formulation gives a decreasing surface area as the precipitation proceeds, it creates a passivation effect. A similar formulation has also been proposed:

$$A_d = A_{d,0} \left( 1 - \left( \frac{V_{d,0} - V_d}{V_{d,0} - V_{d,c}} \right)^n \right) \quad (27)$$

where  $V_{d,c}$  is the threshold volume fraction when the reaction rate becomes negligible, and  $n$  is a fitting parameter. Another formulation is an empirical exponential relationship with a fitting parameter of  $a$ .

$$A = A_0 \exp(-aV_p) \quad (28)$$

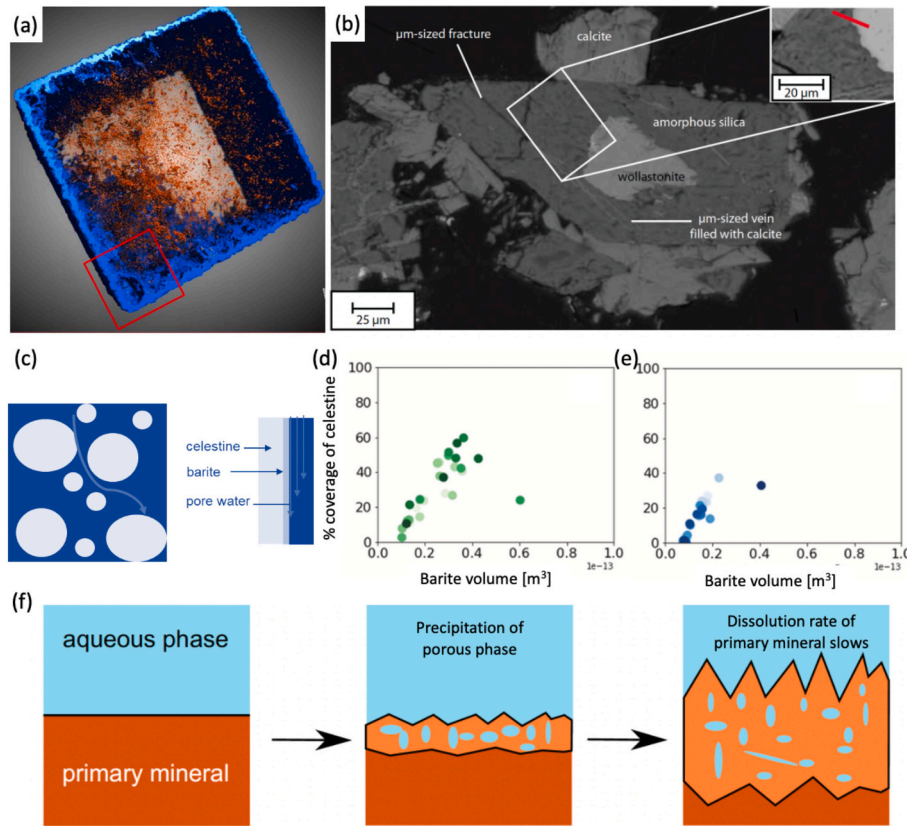
While these relationships have been shown to reproduce experimental observations in certain cases [55,64,128,187]; they fail to capture the entire temporal evolution in some cases and have been shown to be inapplicable in cases with sharp precipitation fronts [128].

One of the reasons for which the passivation effect is difficult to capture via empirical relationships is that the precipitates are porous in many cases. This means that a portion of the surface area of the parent mineral may be retained and accessed via diffusion through the micro-porosity within the precipitates [28,71] (Fig. 11a-b). As a matter of fact, one of the necessary condition for ICDP is the connectivity for transport through the porous secondary mineral phase [141]. So, the change in reactive surface area as approximated in the continuum-scale treatment need to reflect both the reduction in actual surface area and the transport limitation. For example, on a macroscopic scale, a grain may be coated completely, but in presence of internal porosity in the precipitates, the substrate surface may not be 100 % covered on the microscopic scale and the remaining surface area can be accessed through diffusion. The simulations of Hang Deng et al. [33] used a micro-continuum approach that is able to account for the surface area effect and the transport limitation simultaneously, they simulated celestite dissolution and barite precipitation along flow paths within the tortuous pore space between celestite grains (Fig. 11c, left), and the computational domain consisted of a celestite layer on which barite could grow (Fig. 11c, right). They successfully reproduced the experimental observations in [128].

The passivation effect will depend on the surface coverage of the secondary mineral phase. High solution saturation ratios with respect to the secondary mineral lead to elevated nucleation rates, which can result in extensive coverage of the primary mineral surface and thereby limit further dissolution. Hang Deng et al. [33] tested this hypothesis numerically by evaluating the effect of surface coverage of celestite by barite under increasing  $\text{BaCl}_2$  concentrations. Their results showed that higher  $\text{BaCl}_2$  concentrations increased nucleation rates, leading to greater surface coverage of celestite by barite, as illustrated in Fig. 11d (100 mM) compared with Fig. 11e (10 mM). However, the development of a continuous barite rim also depends on other factors, such as the diffusivity of the barite phase and the stochastic nature of nucleation.

Simulations of [12] explicitly consider the fraction of the dissolving mineral coated by the secondary mineral by using the parameter fractional coverage ( $\varphi$ ). The formulation of the specific reactive surface area of the dissolving primary mineral in their model was similar to that of [30] but slightly modified to avoid the need for fitting parameters.





**Fig. 11.** (a) Computed tomography imaging showing the replacement of calcite by cadmium carbonate (blue) and the pores within [71], and (b) SEM image of wollastonite dissolution and precipitation of amorphous silica and calcite at the interface [29]. (c) Conceptual micro-continuum model of celestine dissolution and barite precipitation. (d–e) Numerical results showing the percentage of celestine surface covered by barite at  $\text{BaCl}_2$  concentrations of 100 mM (d) and 10 mM (e) (Hang [33]). (f) Illustration of the development of porosity within the secondary mineral (Simon [36]). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$A^{t+1} = \begin{cases} (1 - \varphi)A_d^t A_p^t < A_d^t \\ 0 A_p^t \geq A_d^t \end{cases} \quad (29a)$$

$$\varphi = \left( \frac{A_p^t / A_d^t}{1 + A_p^t / A_d^t} \right)^\beta \quad (29b)$$

$$\beta = 1 - A_p^t / A_d^t \quad (29c)$$

The extent of the passivation effect also depends on the level and connectivity of microporosity in the precipitation layer. For example, very porous amorphous silica formed following wollastonite dissolution only resulted in a minor passivation effect [29], whereas the silica layer was observed to affect olivine dissolution rate significantly [30]. Emmanuel [36] developed a 1D diffusion boundary layer model to examine how a porous secondary mineral coating alters the dissolution rate of primary minerals under acidic conditions (Fig. 11f). The secondary phase retains connected porosity and allows diffusion through it. The diffusivity is controlled by the ratio between porosity and tortuosity, the increase of which mitigates the passivation effect. The other parameter that was shown to be important is the Biot number, which is equivalent to  $\text{Da}_{\text{II}}$  for the fluid boundary layer. A larger Biot number results in a stronger passivation effect.

The connectivity within the secondary boundary layer may change over time following textural equilibrium [146], or mineral transformation [55]. In the column experiments of brucite dissolution and  $\text{MgCO}_3$  precipitation [55], brucite dissolution was not affected significantly at the early stage as the precipitates are dominated by the poorly

crystalline phase that is more porous and permeable. The amorphous phase was then converted to a more crystalline (flaky/blady) nesquehonite phase, resulting in a stronger passivation effect on and even complete shutdown of brucite dissolution. Varzina et al. [177] carried out a numerical investigation using a lattice Boltzmann framework to explore the effects of crystal shape and packing on dissolution of portlandite and surface precipitation of calcite. In contrast to assuming an effective diffusivity for the precipitate, Varzina's model allowed diffusivity to evolve with the development of the secondary mineral pore structure and incorporated pore-size controlled solubility within nanopores. The study showed that pore size effect has contributed to maintaining the microporosity in the precipitation layer, and calcite layer has diffusivity on the order of  $10^{-13} \text{ m}^2/\text{s}$ . Diffusivity through the calcite layer is the controlling process on portlandite carbonation.

## 5. Advances in experimental and modeling methods

The improvement of our understandings of the precipitation process in (fractured) porous media systems is accompanied by the advances in experimental and numerical methods.

### 5.1. Development of experimental methods

Advances in experimental instrumentation over the past decade have dramatically expanded our ability to study dynamic processes in three dimensions, capturing both spatial and temporal evolution with a level of detail that was previously unattainable. Techniques such as X-ray computed tomography (X-ray CT) now allow dynamic non-invasive visualization of internal structures, enabling researchers to follow

morphological changes and reactive processes as they unfold within complex materials [115,116].

Microfluidic platforms combined with three-dimensional Raman imaging and fluorescence microscopy [132], make it possible to observe chemical reactions and transport phenomena in real time under well-controlled, confined conditions. Atomic force microscopy (AFM) provides complementary insight, allowing the measurement of surface topography and the rates of microscopic precipitation and dissolution, linking molecular- or surface-scale processes to larger-scale system behavior [152,156,185]. In addition, imaging techniques such as magnetic resonance imaging (MRI) [126,187] and positron emission tomography (PET) allow the tracking of fluid flow, solute transport and chemical evolution in porous media, further bridging scales and providing a more complete view of system dynamics. Focused ion beam scanning electron microscopy (FIB-SEM) has opened the door to reconstructing three-dimensional microstructures at nanometer resolution, revealing details of structure and connectivity that were previously inaccessible [186]. By integrating these methods, researchers can now interrogate a range of dynamic processes, from nucleation, crystal growth and changes in fluid flow paths in 3D.

## 5.2. Development of reactive transport modeling

Recent developments in reactive transport modeling of mineral precipitation is featured by improved representations of the precipitation process, the nucleation process and at the pore scale in particular.

The lattice Boltzmann method (LBM) and Smoothed Particle Hydrodynamics (SPH) or particle-based approach in general are among the first that enabled modeling of mineral precipitation at the pore scale [23,73,123,172,173,202]. In these models, the key is to develop a criterion for updating the status of the computational unit following mineral precipitation. In the LBM model, percentage change in the solid volume of a grid cell or a threshold volume fraction was used to convert a fluid cell that allows flow and transport to a solid grid cell, aka the volume of pixel method [73,203]. A similar treatment was adopted in the SPH model, i.e., once the mass of the solid particle changes by a certain amount, the nearest neighboring fluid particle is designated as a solid particle [172,173].

Other commonly used approach to track the migration of mineral-fluid interfaces due to mineral reactions include the Level Set method, the Phase field method, and the Arbitrary Eulerian-Lagrangian (ALE) method [61,86,87,106,107,166,169,199]. In these approaches, the mineral-fluid interface migrates at a local velocity that is related to the mineral reaction rate. While these methods provide an effective way of simulating crystal growth, capturing the initial nucleation process is not straightforward and thus nucleation is not explicitly considered. Additional conceptualization is required to introduce the nucleation event. In the study of Starchenko [169], once the criteria for a nucleation event are satisfied, a nucleus of a pre-defined shape and size is virtually added, and a coverage factor is used to designate a portion of the surface area in the grid cell as reactive, which grows with the nucleus.

The micro-continuum approach provides an alternative that facilitates the modeling of both nucleation and crystal growth [33,35,114,198]. Since local porosity and volume fraction are used to represent the solid phase, fluid phase and the interface, the nucleation events can be readily modeled by updating the mineral volume accordingly to the nucleation rate.

Nucleation rate can be modeled using simple rate laws [40,118,123] or the more rigorous CNT (Section 2.2). Different forms of the CNT has been implemented in several commonly used continuum scale reactive transport models [35,85,170], and in various pore-scale codes [33,137,169,197,198].

Given that nucleation is a result of local fluctuation in stable thermodynamic systems, it is inherently stochastic. Some earlier version of the probabilistic process is realized by converting a neighboring node to a solid node at a prescribed probability [73,202]. Recent studies

modeled the stochastic process by introducing a probability distribution function for the nucleation rate or induction time [33,40,169,199]. For example, in the studies of [33], a random variable that follows a normal distribution ( $x \in N(0, 1)$ ) is used to calculate a stochastic induction time ( $\tau_p$ ) from the induction time given by the theoretical calculation following the CNT ( $\tau_N$ ).

$$\tau_p = \frac{x + n\sigma}{n\sigma} \cdot \tau_N \quad (30)$$

where  $\sigma$  is the standard deviation of the normal distribution and equals to 1 in our case, and  $n = 4$ , capturing 99.9 % of the population in the normal distribution. As a result, the probabilistic induction time falls in  $(0, 2\tau_N]$ . The nucleation rates were calculated from the CNT rate laws (Eqs. (6)–(11)) and adjusted according to the probabilistic induction time.

$$J_p = J \cdot \frac{\tau_N}{\tau_p} \quad (31)$$

## 6. Frontier research on mineral precipitation in fractured porous media

### 6.1. Outstanding challenges of experiments and modeling

Some remaining challenges of the pore-scale modeling of the precipitation process includes the considerations of the pore size effect, the non-classical pathways, crystal morphology, and mineral transformation. The study of Starchenko [169] considered the impact of local curvature on reaction rate, by defining an effective equilibrium concentration based on local curvature, grain size and interfacial tension following the Gibbs-Thomson equation, such that larger local curvature results in higher effective equilibrium concentration and thus lower thermodynamic driving force for both the nucleation and growth process. It provides an example of accounting for local curvature and surface tension effect, that may be used to explicitly model the pore-size effect or to assign different reaction rates based on the crystal facet to produce certain crystal morphology (Fengchang [198]). A phase-field method based on energy minimization provides another approach to model crystal habits [166].

One outstanding numerical challenge is related to the possibility of overshoot or the break of mass conservation. When transport and reactions are solved sequentially, large time steps may result in precipitation amount that exceeds the available solutes in the solutions. Time steps must be chosen carefully. However, the computational cost, especially considering fine spatial scales necessary to capture e.g., local curvature, can be prohibiting. The development of hybrid modeling approach, and adaptive meshing and time stepping are in need.

Co-design of experimental and numerical studies are desired. The development of numerical models requires the availability of benchmark experiments, while numerical tools are key for isolating complex mechanisms that is difficult under experimental conditions. The use of artificial intelligence provides an opportunity to further the advances of both numerical and experimental studies of precipitation in porous media. For example, by leveraging surrogate models trained on geochemical and hydrodynamics data from process-based modeling and experimental observations to identify critical events and predict system evolution, experimental platforms can be tuned dynamically to investigate specific processes with unprecedented precision, effectively creating “intelligent” experiments for investigating mineral precipitation in porous media [139]. Moreover, the rapid integration of artificial intelligence and automated workflows into experimental design and analysis has the potential to redefine how experiments are conceived and conducted, moving beyond the classical paradigm of post data processing toward a framework in which experiments can be actively guided and optimized in real time [158].

## 6.2. Solid solution precipitation

Beyond the formation of pure mineral phases, solid solutions can also precipitate in porous media, playing an important role in controlling the mobility of trace and toxic elements in subsurface environments. Co-precipitation processes are widely employed in engineered remediation strategies and engineered system to immobilize hazardous species. For example, the formation of  $(\text{Ba,Ra})\text{SO}_4$  solid which act as effective scavengers for the long-lived radionuclide  $^{226}\text{Ra}$  (half-life of 1300 years), is known to precipitate in diverse contexts, including geothermal systems, hydraulic fracturing, uranium mining operations, and nuclear waste repositories [127]. While atomistic simulations can provide insights into the mixing behavior and thermodynamic properties of these solid solutions [181–183], incorporating such behavior into geochemical or reactive transport models remains challenging. Currently, GEMS is one of the few simulation tools capable of accurately modeling non-ideal solid solutions. Most other tools require explicit specification of discrete solid solution phases [90]. The composition of solid solutions can evolve over time due to thermodynamic equilibration, making an accurate formulation of solid solutions essential for predicting aqueous radionuclide concentrations e.g.,  $^{226}\text{Ra}$  in the near field of nuclear waste repositories.

Oscillatory zoning is a common feature of binary solid solutions with end-members of widely differing solubilities, such as  $(\text{Ba,Sr})\text{SO}_4$  and  $\text{Ba}(\text{SO}_4,\text{CrO}_4)$ . In diffusion-limited systems, including tight rock formations or engineered barriers in nuclear waste repositories, local variations in aqueous composition can induce sequential precipitation of compositionally distinct layers. Experimental studies have demonstrated oscillatory zoning in  $(\text{Ba,Sr})\text{SO}_4$  solid solution under controlled counter-diffusion conditions (Fig. 12a). In one setup, a porous silica gel matrix was placed between two solution reservoirs: one containing a mixture of  $\text{BaCl}_2$  and  $\text{SrCl}_2$ , and the other containing  $\text{Na}_2\text{SO}_4$ . Slow diffusion of solutes through the gel induced the sequential crystallization of zoned  $(\text{Ba,Sr})\text{SO}_4$  crystals, with alternating Ba- and Sr-enriched layers. This behavior arises from the alternating depletion of Ba and Sr in the pore solution adjacent to the crystal surfaces, reflecting a dynamic interplay between local supersaturation and crystal growth. Although the lower solubility end-member ( $\text{BaSO}_4$ ) would be expected to precipitate first under equilibrium conditions, Sr-enriched solid solutions were observed to nucleate initially. This unexpected sequence was attributed to non-equilibrium nucleation kinetics:  $\text{SrSO}_4$  requires a much lower supersaturation ( $\sim 100$ ) than  $\text{BaSO}_4$  ( $\sim 10,000$ ) to nucleate, allowing Sr-enriched layers to form first. Recent investigations combining microfluidics and in-situ Raman spectroscopy have further refined our understanding, showing that oscillatory zoning is not solely a consequence of diffusion-limited solute transport. Instead, the phenomenon is also influenced by the nucleation rates of individual stoichiometric compositions. Even when a solid solution is the thermodynamically most stable phase (Fig. 12b), its precipitation may be kinetically hindered if other phases possess higher nucleation rates under the prevailing conditions

(Fig. 12c). These findings highlight the importance of integrating both thermodynamic and kinetic factors when modeling co-precipitation and zonation in porous media.

## 6.3. Precipitation induced cracking

In the precipitation process, the growth of mineral crystals can exert stress on the surrounding porous matrix, which is known as the crystallization pressure [159]. The crystallization pressure depends on the volume change caused by the precipitating mineral, and the precipitation rate, which is determined by factors such as supersaturation [77,159]. When the capillary pressure exceeds the tensile strength of the porous material, cracks may nucleate and propagate [77,192]. At crystallographic defects or features such as etch pits caused by dissolution reactions, crack propagation may require a lower stress and is thus more susceptible to precipitation induced cracking [63,71]. This implies that precipitation induced cracking may be highly relevant in  $\text{CO}_2$  carbonation systems, especially when the precipitation rate is fast [98].

Precipitation induced cracking is an important process that alters fractured porous media, as it creates porosity, permeability and transport pathways, as well as new surface areas. It has been recognized as a mechanism that sustains transport through the secondary precipitation layer, a necessary condition for interface coupled dissolution and precipitation [5,71]. At the larger spatial scale, reactive cracking is also considered as a key process that maintains permeability, which then allows fluid flow and supply of reactants and thus enables continued reactions [77]. However, quantitative analyses that consider the feedback between the geochemical and mechanical processes are still in short.

## 6.4. Upscaling

Large-scale modeling of precipitation-driven, or generally reaction-driven, alteration of fractured porous media – which is necessary for investigations of many natural and engineered systems – requires the use of constitutive relations. As discussed above, the constitutive relations do not have a universal form and are highly dependent on pore-scale and sub-pore scale processes and properties. Integrating such fine-scale information into the predictions of long-term alteration of fractured porous media is crucial. Fig. 13 provides an illustration of the relevant properties and processes that were discussed in the paper and are involved in precipitation in fractured porous media, and their connections at different scales.

The first challenge of upscaling lies in our lack of knowledge of some of the important fine-scale processes and properties, and their interactions (e.g., Fig. 13). As such, these effects are not considered in any of the constitutive relationships and are lost in large scale models. For example, nano-scale heterogeneity in surface properties (e.g., reactivity) of the substrate [3], and property change of water/solutions under nano-confinement exemplify relevant nano-scale effects. Quantifying these

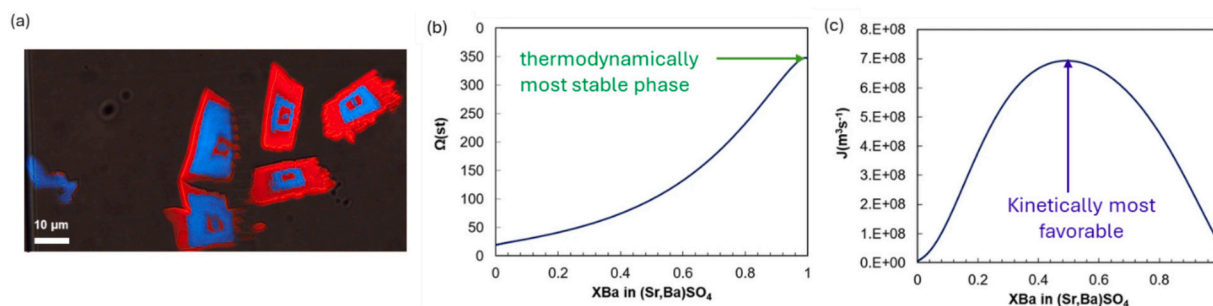
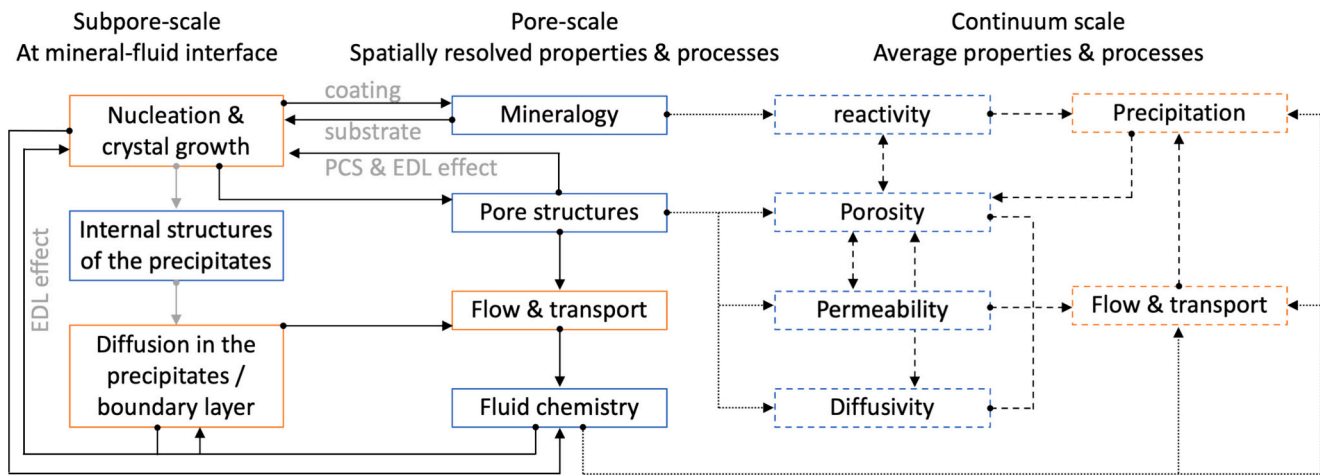


Fig. 12. Oscillatory zoning from [130,133] (a) Alternating layer of Barium enriched (blue) and strontium enriched (red) solid solution. (b) Saturation function i.e. the specific saturation state of the solid solution series of  $(\text{Ba}_x\text{Sr}_{1-x})\text{SO}_4$  for a given aqueous solution composition and associated nucleation rate of the  $(\text{Ba}_x\text{Sr}_{1-x})\text{SO}_4$  solid solution series. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)





**Fig. 13.** Illustration of the precipitation in fractured porous media at different scales (with solid lines corresponding to pore-scale and sub-pore scale, and dashed lines corresponding to continuum scale), and some key properties (blue box) and processes (orange box). Dotted arrows denote the upscaling process, which in current models rely heavily on averaging and homogenization. Gray arrows and texts highlight subpore-scale effects that are not well understood. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

effects requires upscaling molecular level information into macroscopic rate laws, which remains challenging [167]. While molecular dynamics simulations and advances in spectroscopic and microscopic techniques have enabled improved understanding of the aforementioned nanoscale effects, our comprehension of nucleation and crystal growth at the nanoscale remains limited [148]. For another example, formation and evolution of the internal structures of mineral precipitates are also key in controlling sub-pore scale transport, which translates into macroscopic reactivity, and thus the overall evolution of bulk properties. While high-fidelity model that resolves the internal structure of secondary mineral layer [177] and micro-continuum treatment that also involves homogenization and simplification of the precipitation layer [33,36] have been adopted to account for the porous nature of the precipitates, our knowledge about the controls on the level and connectivity of the microporosity within the precipitates is deficient for developing quantitative predictions.

Another challenge of upscaling is that flow, diffusion, reactions and structural evolution are intricately coupled at the fine-scale, and using separate constitute relationships (as shown by the dashed double arrows in Fig. 13) that upscale their macroscopic effects may be insufficient and coordinated upscaling of the coupled processes may be needed. For example, pore-scale hydrodynamics and oscillatory behaviors due to feedbacks with pore structure changes demonstrate the difficulty of upscaling pore-scale dynamics into constitutive relations such as the porosity-permeability relationship. To address this challenge, one strategy is to include additional parameters in formulating the constitutive relationships to account for complex chemical-physical interactions. For instance, in the numerical study of Ahkami et al. [4], the different precipitation regimes in fractured porous media led to different porosity-permeability relationships. The authors proposed a formulation that is able to account for the impacts of precipitation patterns by introducing a function of the concentration field. While this study focuses on a highly simplified single-species mineral precipitation, and thus the applicability of the proposed formulation in more general cases needs to be examined, it illuminates on a possible path toward considering different reactive transport regimes and the resulting precipitation patterns and thus porosity-permeability relationships. This is also consistent with the previous discussion of using a scale separation parameter that incorporates concentration gradient to determine the applicability of macroscopic relationships/models [8]. An alternative approach as proposed by N. I. Prasianakis et al. [138] is to create a look-up table based on pore-scale simulations and a full coupling between the pore-scale model with the macroscopic model for upscaling. Machine

learning based approach has also been adopted and provides an alternative framework [78]. However, this approach requires abundant experimental data, and given the complex dependence of mineral precipitation on reactive transport conditions and the geochemical systems, algorithms trained for one geochemical and hydrodynamic system may not be broadly applicable.

One more challenge of upscaling is the need to cover a wide range of scales in many cases, from nano-/micro-scale to field-scale (~km). In other words, the domain size of fine scale studies used to develop constitutive relations may be far smaller than the grid cell size of the large-scale studies in which averaged properties (e.g., Fig. 13 dotted arrows) and the constitutive relations are used. The multi-stage upscaling approach as proposed by M Masoudi et al. [97] provides a potential solution. In their framework, a stochastic porosity-permeability relationship was used to account for the impacts of probabilistic nucleation. In this multi-stage upscaling approach, pore-scale simulations were performed to generate a probability distribution for the key parameters of the porosity-permeability relationship (i.e., the exponent of the power law relation). In the core-scale study, the porosity-permeability relationship used in each grid cell samples from the probability distribution. Upscaling from the core-scale to the application scale then repeats the same process. Another manifestation of this multi-scale challenge in upscaling is fractured porous media, as fractures and pores typically have significantly different characteristic length scales. While the underlying fundamental controls on precipitation process are the same in fractures versus porous media, flow and transport in fractures tend to create different local reactive transport conditions compared to the porous media, and thus different forms of constitutive relationships may be needed. Also, in fractured porous media, accounting for mass transfer between fractures and porous matrix and their evolution following mineral precipitation in large scale modeling (e.g., multi-continua interacting models) are not straightforward. Resolving some of the numerical challenges discussed in 6.1 may help address the upscaling of precipitation in fractured porous media as well.

## 7. Concluding remarks

Precipitation in porous media is a widely observed phenomenon in natural and engineered systems, and predictive understanding of this process is key for accurate assessment and engineering designs of these systems.

Mineral precipitation involves nucleation and crystal growth, and

cannot always be treated similarly as mineral dissolution. The precipitation reactions may have complex dependencies on the substrate, including the mineralogy and topography, and on pore sizes as (nano) confinement can affect reaction thermodynamics and kinetics. The extent and spatial distributions of mineral precipitates are also largely shaped by local reactive transport conditions.

For systems involving the injections of supersaturated solutions, precipitation is more spatially heterogeneous under transport controlled conditions, i.e., precipitation rate is fast compared to transport rate. In contrast, under surface reaction controlled conditions, precipitation tends to be more spatially homogeneous, and spatial gradients may develop following the drop of SI at a larger spatial scale. The presence of fractures amplify the intrinsic spatial heterogeneity in the media. Mineral precipitation can be localized at the fracture surface when the precipitation rate is fast relative to transport into the matrix, and can penetrate into the matrix when transport into the matrix is fast enough.

Precipitation induced by mixing of two solutions is featured by a thin mineral band along the mixing line. Thus, the boundary conditions will dictate the hydrodynamics of mixing and precipitation dynamics. Typically, more heterogeneous pore structures would result in more tortuous mixing line and precipitation band, and the width of the precipitation band is smaller at faster precipitation rate.

In coupled dissolution and precipitation systems, i.e., dissolution is necessary to bring about precipitation, the dependence of mineral precipitation on reactive transport conditions is more complicated. At slow precipitation rates, the dissolution and precipitation reactions are more likely to be spatially decoupled. Minerals dissolve more substantially in regions with fast transport, i.e., advection-dominated regions, whereas secondary mineral precipitates primarily in diffusion-dominated regions. With faster precipitation rates, hydrodynamics boundary layer at the interface of the parent mineral phase may be enough to create conditions for interface-coupled dissolution and precipitation, in which dissolution and precipitation are spatially co-located.

Mineral precipitation can cause narrowing of pores and pore throats and clogging of transport pathways. At the continuum scale, it manifests as reductions in porosity, permeability and diffusivity. Traditional porosity-permeability relationships (e.g., the Kozeny Carman relationship and power law relationship) and porosity-diffusivity relationships (e.g., Archie's law) may fail in capturing the clogging effects. Introducing a critical porosity that is associated with the point of clogging – as is the case in the Verma-Pruess porosity-permeability relationship and the extended Archie's law – has been shown to improve the predictions of permeability and diffusivity in precipitation dominated systems. However, this practice doesn't provide a silver bullet, as it does not capture all the influencing pore-scale dynamics, especially those involve multiphase flow and anisotropy.

The resulting impacts of mineral precipitation on reactivity is primarily reflected by changes in surface areas, which may be quite different depending on whether they are for the primary/dissolving mineral phase or the secondary/precipitating mineral phase. While surface area of the precipitating phase may increase following roughening, the surface area of the primary mineral phases typically decreases as it is coated by the precipitating phase. However, this approach does not explicitly consider transport through the precipitates that may be porous, as the internal porosity and transport properties of the precipitates are difficult to quantify.

Advances in experimental and modeling studies of precipitation in fractured porous media are featured by the development and popularization of CT scanners, microfluidic cells and the co-practicing of multiple characterization tools, and by the more rigorous representation of nucleation and crystal growth at the pore scale. While they have significantly enhanced our understanding of precipitation in fractured porous, challenges remain. Some of the research directions that call for dedicated future investigations include improved quantitative descriptions of non-classical nucleation and formation of solid solution. Furthermore, precipitation-induced cracking warrants more studies, for

the coupling between geochemical and geomechanical processes is a mechanism that may cause changes in the bulk properties in the opposite direction. Ultimately, being able to upscale fine scale information, ranging from molecular nano-scale to pore scale, is critical for developing robust macroscopic relations and predictions.

## CRediT authorship contribution statement

**Hang Deng:** Writing – review & editing, Writing – original draft, Visualization, Funding acquisition, Conceptualization. **Jenna Poonosamy:** Writing – review & editing, Writing – original draft, Visualization, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

No data was used for the research described in the article.

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